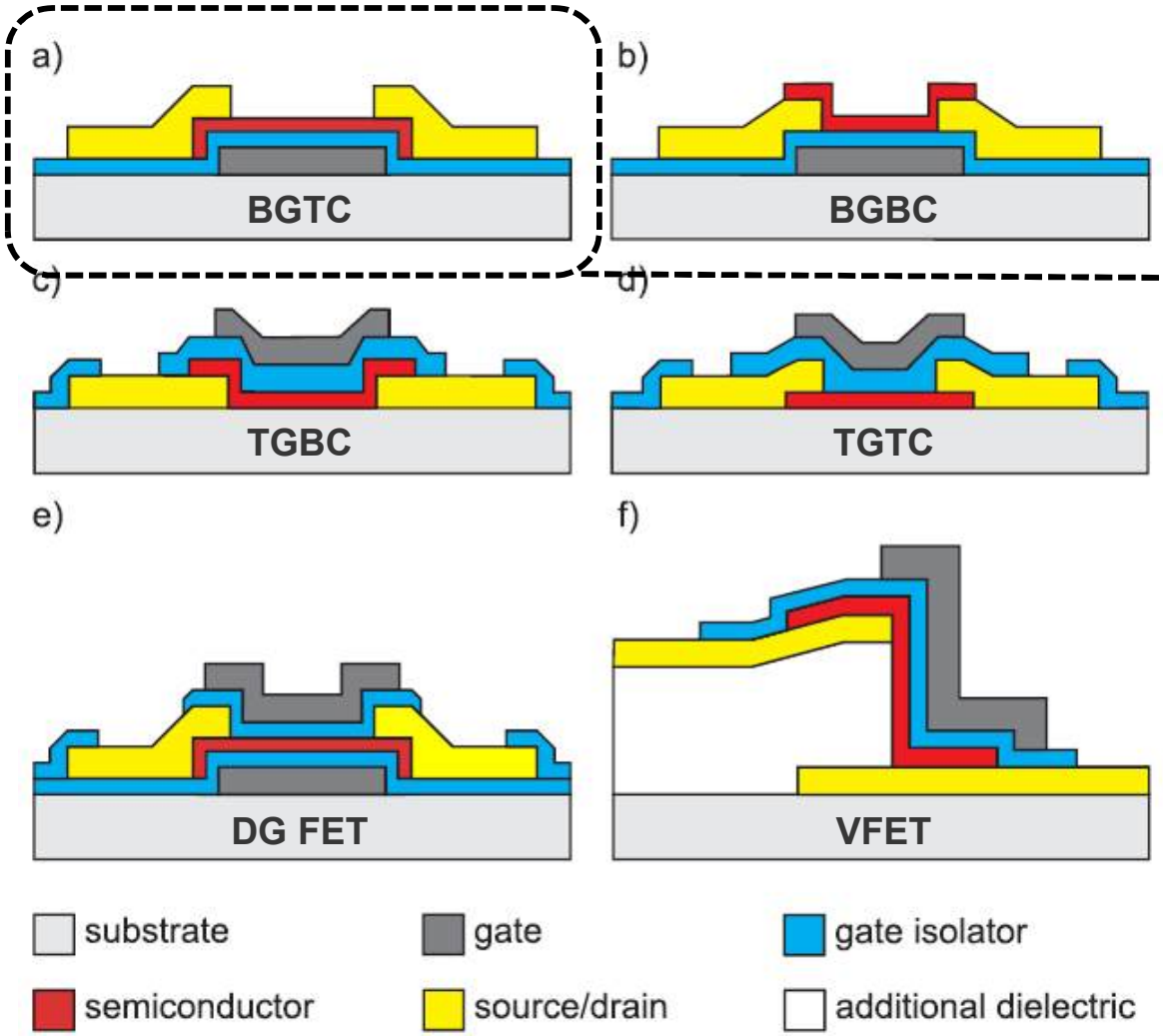
Fabrication of Oxide TFTs

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Molecular Engineering for Next Generation
Nano-Electronics Laboratory,
School of Electronic and Electrical Engineering,
Kyungpook National University,
Daegu, Republic of Korea

1. Oxide TFTs

1.1 Configuration



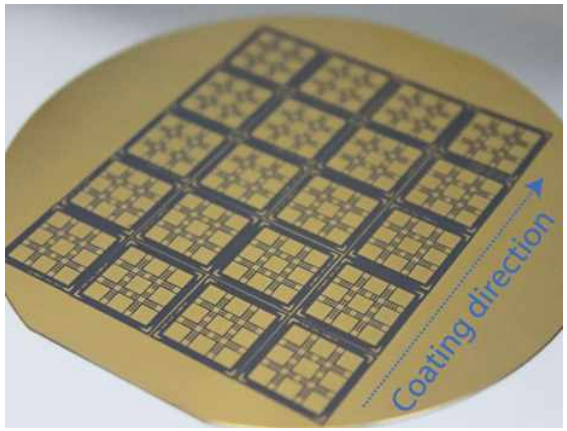
일반적으로 amorphous Si TFT 와 oxide TFT에서 BGTC 구조를 사용.

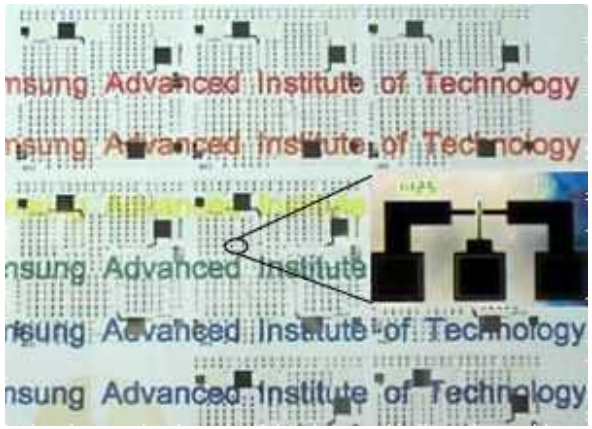
본 실험에서 또한, BGTC 구조로 oxide TFT를 제작할 예정.

1. Oxide TFTs

1.1 Materials

TFTs	Rigid & opaque	Transparent	Flexible
Substrate	Highly doped Si	Glass	PI, PET, PEN
Gate	Highly doped Si	ITO	ITO
Insulator	SiO_2	SiN, SiO_2, Metal-oxide, Polymer dielectrics	Metal-oxide, Polymer dielectrics
Semiconductor	In_2O_3, IZO, IGZO, etc.	In_2O_3, IZO, IGZO, etc.	In_2O_3, IZO, IGZO, etc.
Source/drain	Metal (Al, Au)	Metal, ITO	Metal, ITO





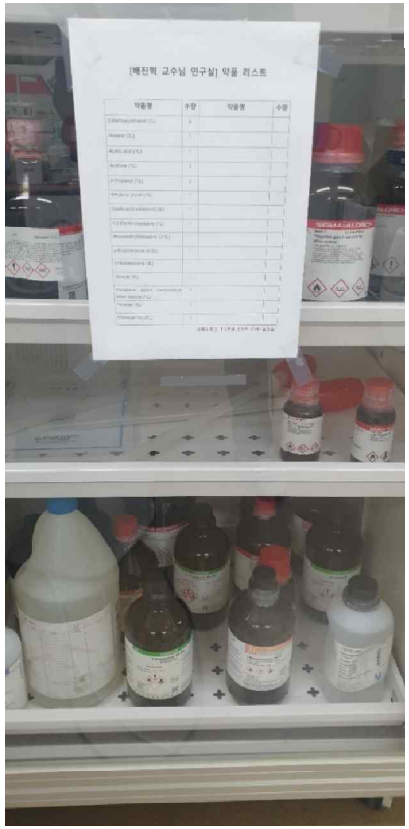


2. Fabrication Process

2.1 Materials Preparation : Oxide precursor solution

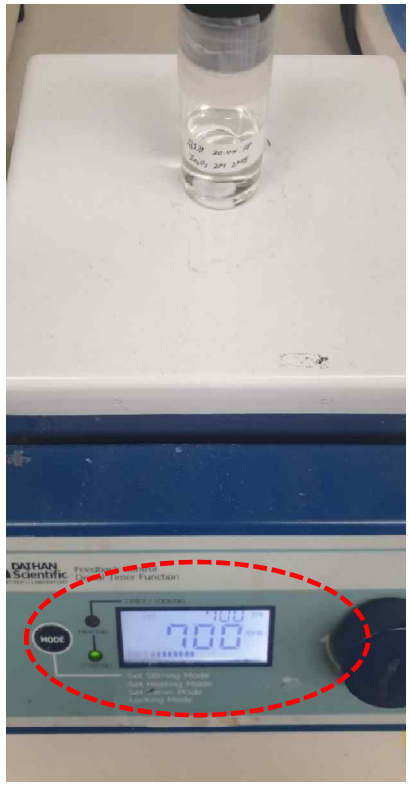


Solute



Solvent

0.1M In_2O_3 semiconductor solution



Solution

(Stirring at 700 rpm, 12 h)

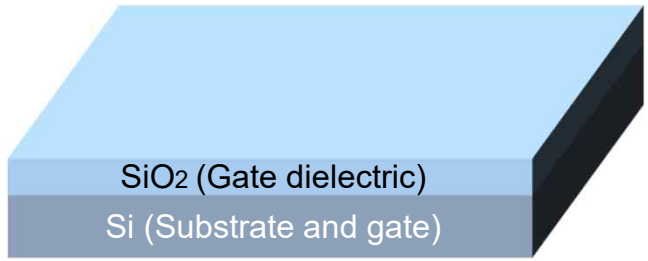
2. Fabrication Process

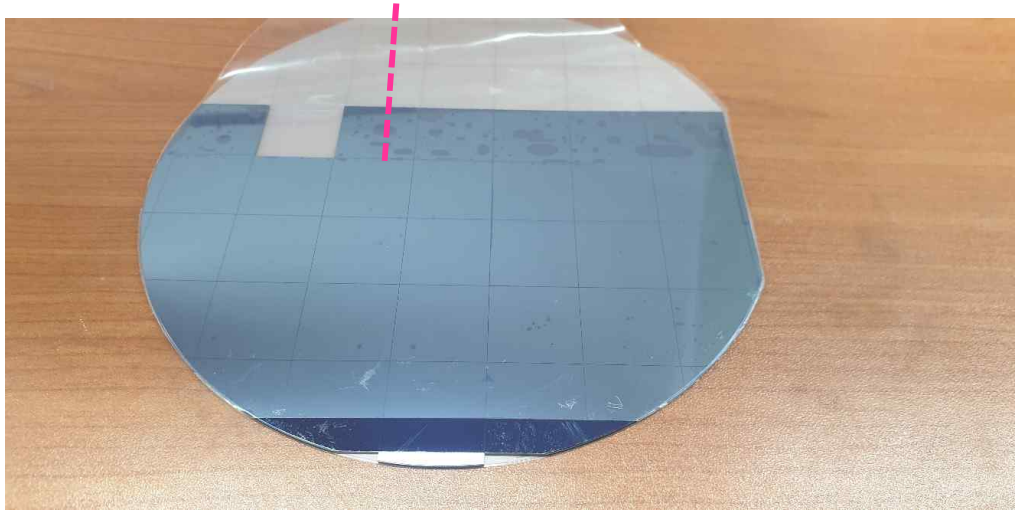
2.1 Materials Preparation : Substrate





Heavily p-type boron doped Si/SiO_2 wafer





2. Fabrication Process

2.2 Cleaning

1) Sonication
1-1) In acetone for 10 min
1-2) In isopropyl alcohol (IPA) for 10 min
1-3) In deionized (DI) water for 10 min



Acetone



IPA



DI water



Sonicator



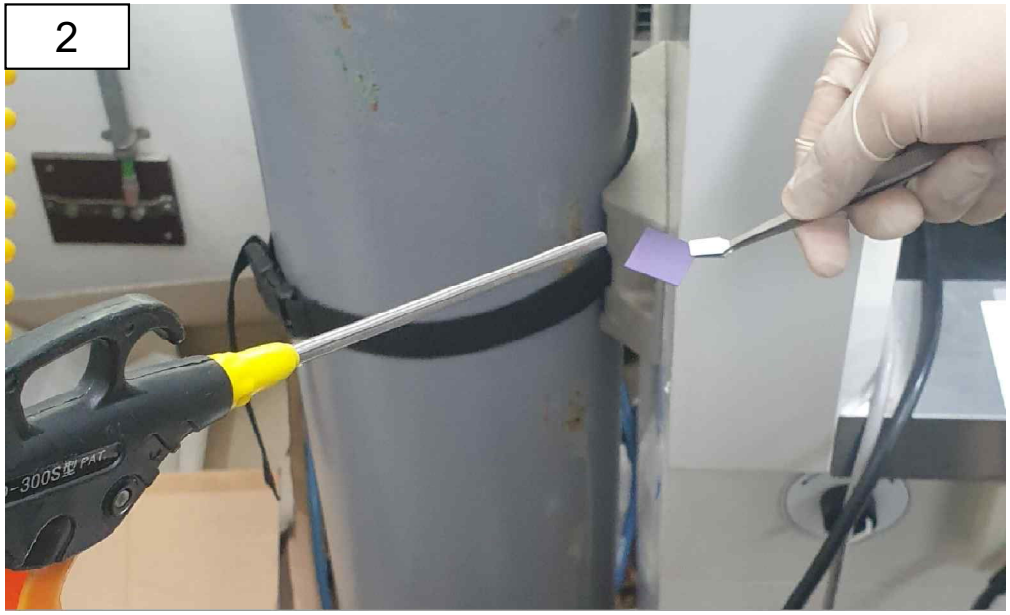
Sonication

2. Fabrication Process

2.2 Cleaning

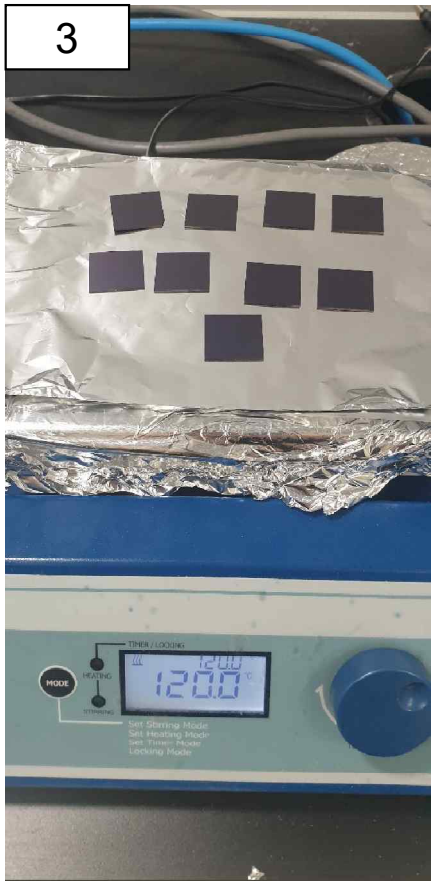
2) Stream of nitrogen gas
3) Heated on a hot plate at 120℃ for 1 h

2



물기 제거

3



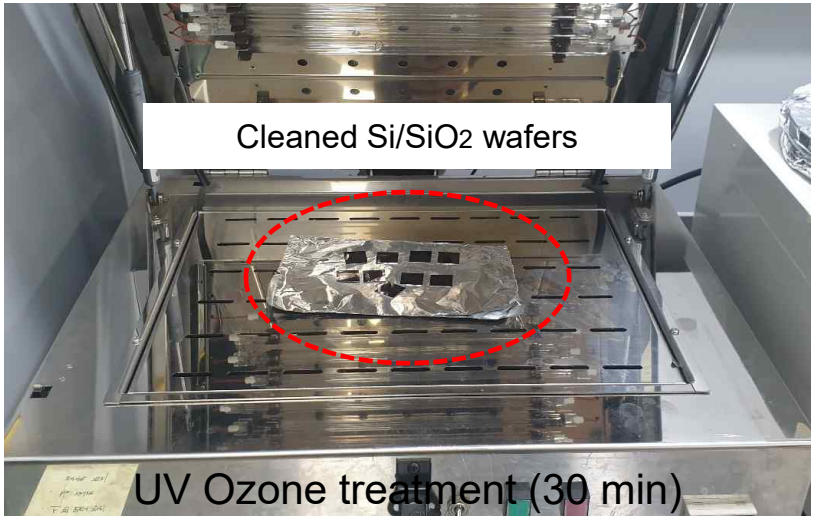
잔여 수분 제거

2. Fabrication Process

2.3 UVO (or O_2 Plasma) Treatment



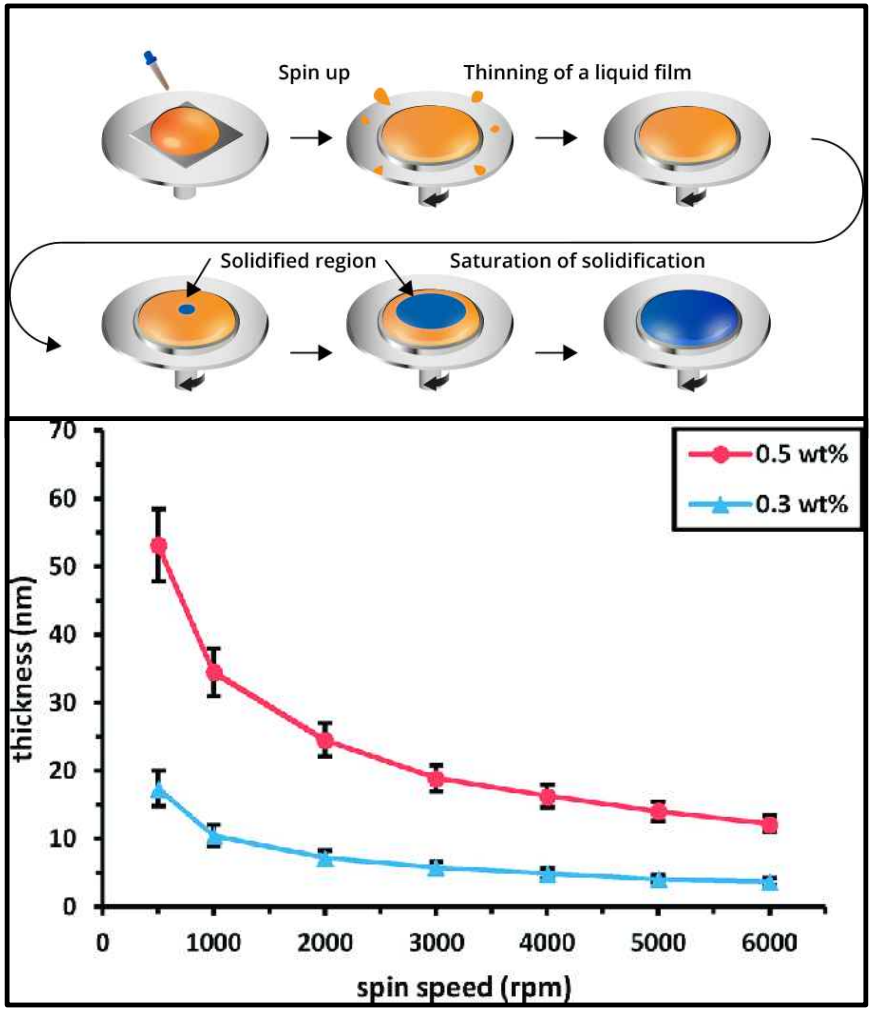
UV Ozone cleaner



UV Ozone treatment
1) 유기 잔여물을 제거하여 기판 세정효과를 가져옴
2) 오존과 산소 형성을 통해 친수성 기판 표면을 형성해주어, 향후 기판에 반도체 용액이 잘 도포되게 도와줌

2. Fabrication Process

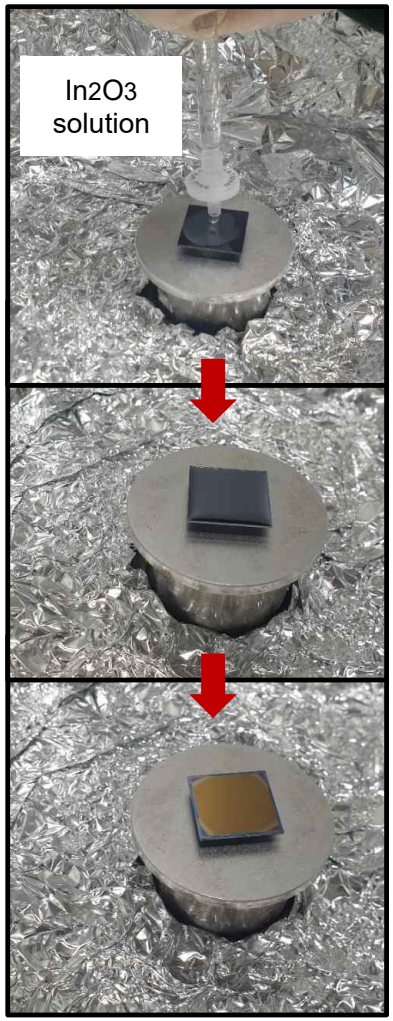
2.4 Solution Coating: Spin Coating





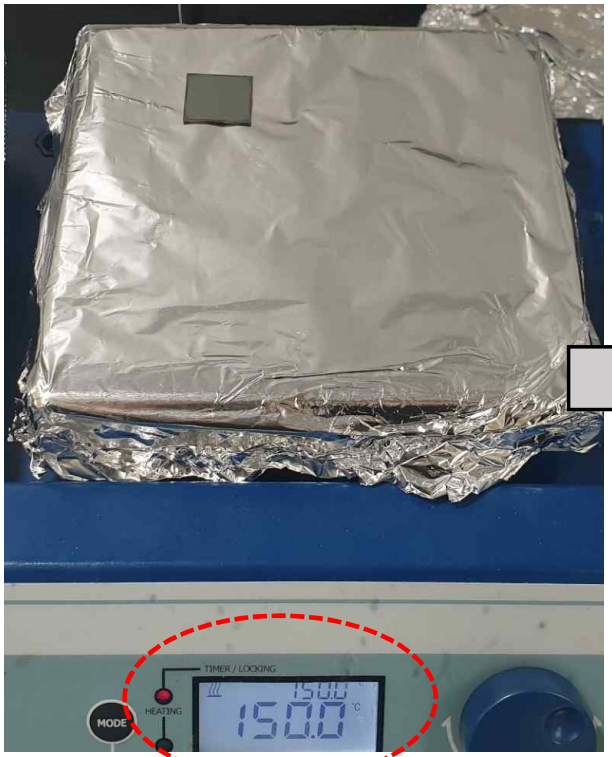
Spin coater

In_2O_3 solution spin-coating:
3000 rpm, 20 sec



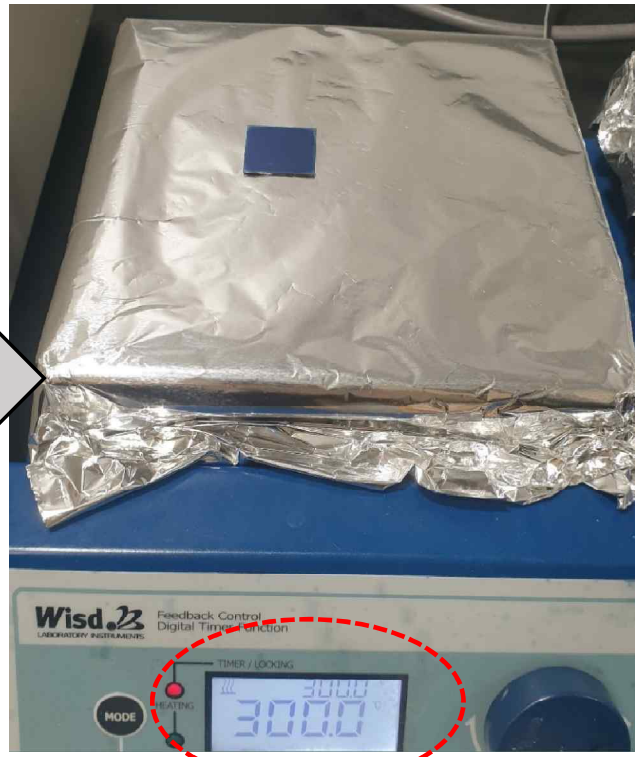
2. Fabrication Process

2.5 Solidification and Densification (Activation)



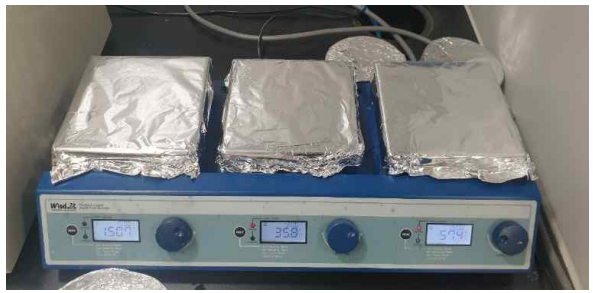
Pre-annealing of In_2O_3 thin-film

150℃ for 5 min
도포된 용액의 고체화 (solidification)



Post-annealing of In_2O_3 thin-film

300℃ for 1 h
반도체 박막의 활성화 (activation)



Hot plate (substrate heating)

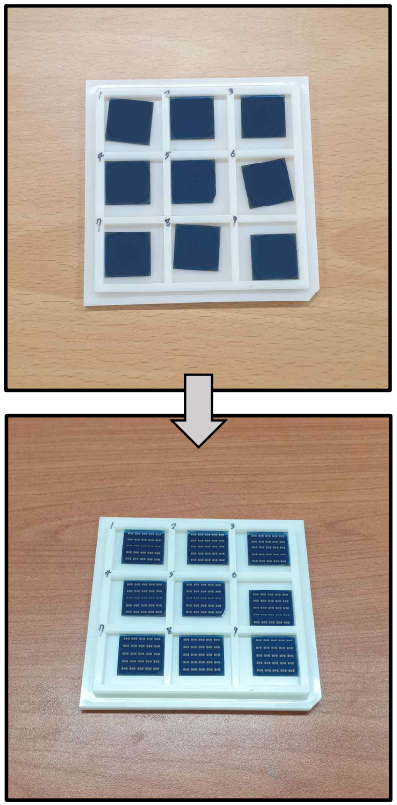


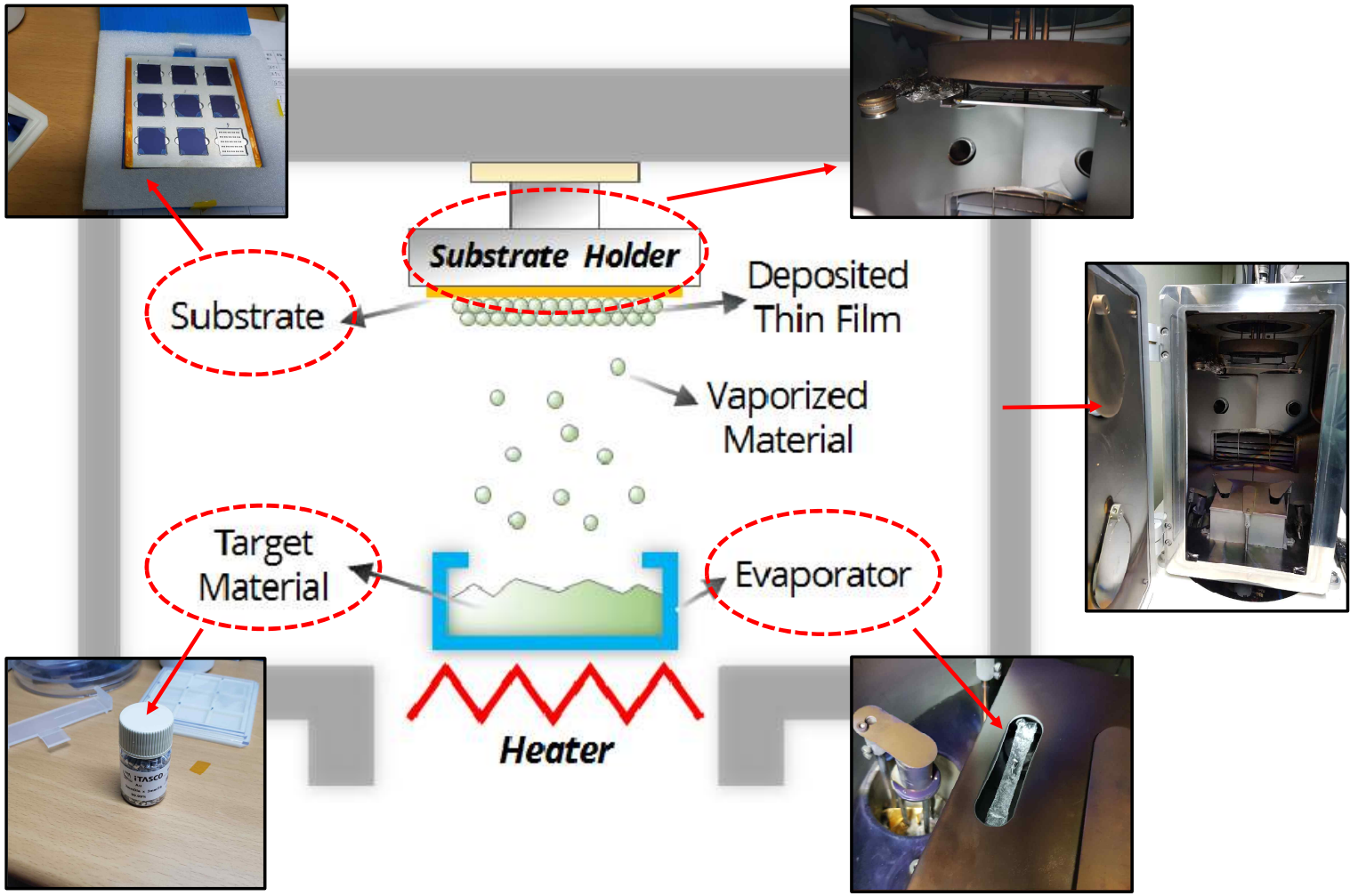
Tube furnace (air heating+gas flow)

2. Fabrication Process

2.6 Source/Drain Electrodes Deposition

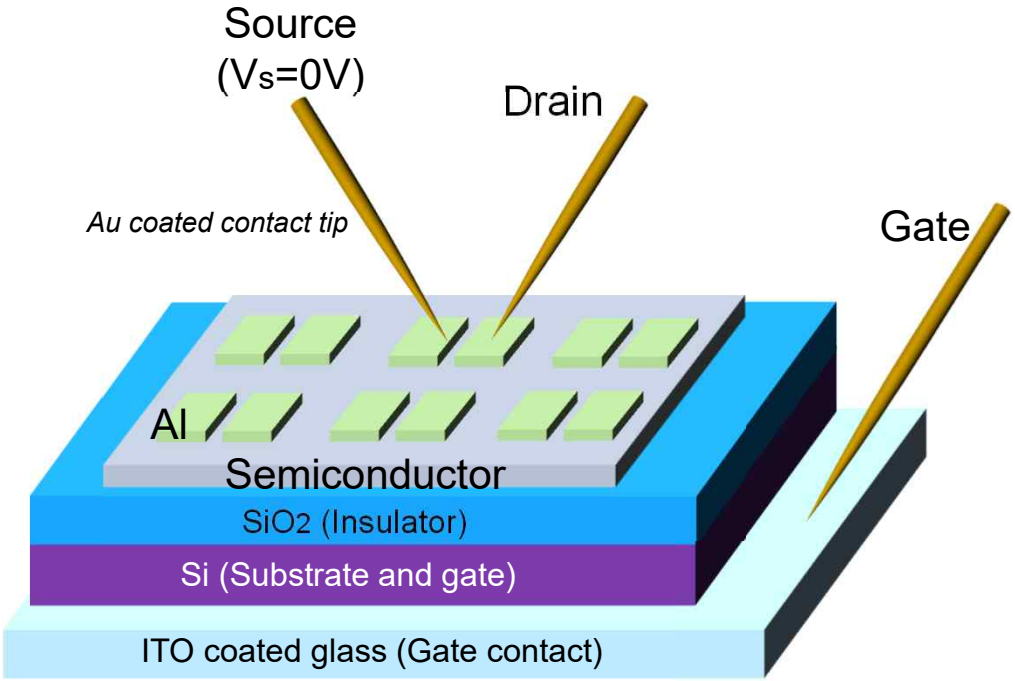
Thermal evaporation
- Shadow mask
- Pressure: 10E-06 torr
- Evaporation rate: 0.1 nm/s
- Al thickness: 50 nm

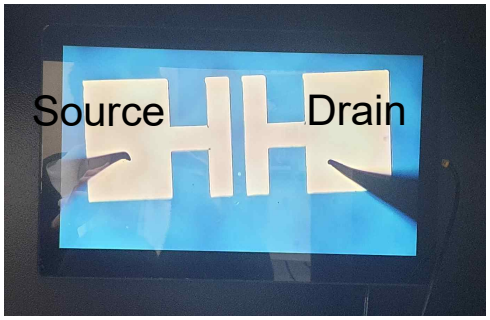




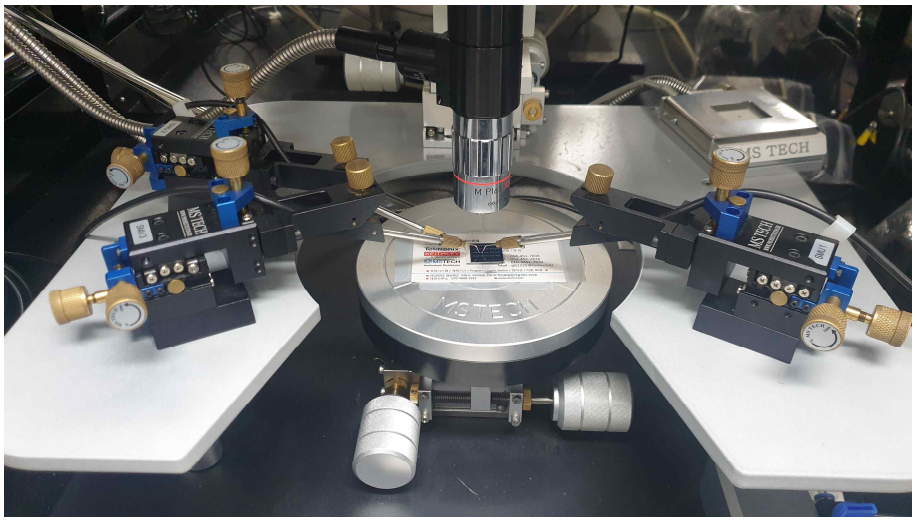
3. Device Measurement

3.1 Probe Station









Silicon 기판 뒷면의 얇은 자연 산화막(native silicon oxide)을 다이아몬드 칼로 긁어낸 후, silver paste를 바르고 ITO가 코팅된 glass를 붙여 gate contact.
또는 Silicon 기판 상부의 절연체를 긁어낸 후, gate contact.

3. Device Measurement

3.2 Obtaining Raw Datas

Raw data of transfer curve

	1-VG	2-IG	3-VD	4-ID	5-|ID|	6-sqrt|ID|
1	-30.0000	-4.3325e-7	40.0000	2.1705e-7	2.1705e-7	4.6589e-4
2	-29.0000	-2.1314e-7	40.0000	1.4173e-7	1.4173e-7	3.7648e-4
3	-28.0000	-1.1400e-7	40.0000	5.2505e-8	5.2505e-8	2.2914e-4
4	-27.0000	-4.1587e-8	40.0000	3.1592e-8	3.1592e-8	1.7774e-4
5	-26.0000	-3.2616e-8	40.0000	2.5920e-8	2.5920e-8	1.6100e-4
6	-25.0000	-2.6971e-8	40.0000	2.2008e-8	2.2008e-8	1.4835e-4
7	-24.0000	-2.2757e-8	40.0000	1.8980e-8	1.8980e-8	1.3777e-4
8	-23.0000	-1.9595e-8	40.0000	1.6635e-8	1.6635e-8	1.2898e-4
9	-22.0000	-1.6971e-8	40.0000	1.4699e-8	1.4699e-8	1.2124e-4
10	-21.0000	-1.4937e-8	40.0000	1.3190e-8	1.3190e-8	1.1485e-4
11	-20.0000	-1.3291e-8	40.0000	1.1863e-8	1.1863e-8	1.0892e-4
12	-19.0000	-1.1669e-8	40.0000	1.0773e-8	1.0773e-8	1.0379e-4
13	-18.0000	-1.0515e-8	40.0000	7.8758e-9	7.8758e-9	8.8746e-5
14	-17.0000	-6.7395e-9	40.0000	5.5452e-9	5.5452e-9	7.4466e-5
15	-16.0000	-5.2619e-9	40.0000	4.4492e-9	4.4492e-9	6.6702e-5
16	-15.0000	-4.2671e-9	40.0000	3.6763e-9	3.6763e-9	6.0632e-5
17	-14.0000	-3.5847e-9	40.0000	3.1024e-9	3.1024e-9	5.5699e-5
18	-13.0000	-3.0406e-9	40.0000	2.6601e-9	2.6601e-9	5.1576e-5
19	-12.0000	-2.6069e-9	40.0000	2.3129e-9	2.3129e-9	4.8093e-5
20	-11.0000	-2.2528e-9	40.0000	2.0337e-9	2.0337e-9	4.5097e-5
21	-10.0000	-1.9514e-9	40.0000	1.8038e-9	1.8038e-9	4.2471e-5
22	-9.0000	-1.7406e-9	40.0000	1.6138e-9	1.6138e-9	4.0172e-5
23	-8.0000	-1.5421e-9	40.0000	1.4481e-9	1.4481e-9	3.8054e-5
24	-7.0000	-1.3179e-9	40.0000	1.3120e-9	1.3120e-9	3.6221e-5
25	-6.0000	-1.0356e-9	40.0000	1.1956e-9	1.1956e-9	3.4577e-5
26	-5.0000	-7.6331e-10	40.0000	4.6243e-10	4.6243e-10	2.1504e-5
27	-4.0000	-4.2578e-10	40.0000	3.1835e-10	3.1835e-10	1.7842e-5
28	-3.0000	-3.1573e-10	40.0000	2.4284e-10	2.4284e-10	1.5583e-5
29	-2.0000	-2.3152e-10	40.0000	2.0351e-10	2.0351e-10	1.4266e-5
30	-1.0000	-1.7085e-10	40.0000	1.7589e-10	1.7589e-10	1.3262e-5
31	0.0000	-1.4495e-10	40.0000	1.5311e-10	1.5311e-10	1.2374e-5
32	1.0000	-1.3059e-10	40.0000	1.5034e-10	1.5034e-10	1.2261e-5
33	2.0000	9.4849e-10	40.0000	7.6295e-9	7.6295e-9	8.7347e-5
34	3.0000	8.9492e-9	40.0000	4.8480e-8	4.8480e-8	2.2018e-4
35	4.0000	1.9337e-8	40.0000	1.3703e-7	1.3703e-7	3.7017e-4
36	5.0000	6.0785e-8	40.0000	2.9943e-7	2.9943e-7	5.4720e-4
37	6.0000	9.9578e-8	40.0000	5.3963e-7	5.3963e-7	7.3460e-4
38	7.0000	1.2892e-7	40.0000	8.6022e-7	8.6022e-7	9.2748e-4
39	8.0000	2.2419e-7	40.0000	1.2851e-6	1.2851e-6	1.1336e-3
40	9.0000	2.9964e-7	40.0000	1.8163e-6	1.8163e-6	1.3477e-3
41	10.0000	3.9134e-7	40.0000	2.4508e-6	2.4508e-6	1.5655e-3
42	11.0000	4.8508e-7	40.0000	3.1862e-6	3.1862e-6	1.7850e-3
43	12.0000	6.0078e-7	40.0000	4.0338e-6	4.0338e-6	2.0084e-3
44	13.0000	7.2300e-7	40.0000	4.9813e-6	4.9813e-6	2.2319e-3
45	14.0000	8.6856e-7	40.0000	6.0584e-6	6.0584e-6	2.4614e-3
46	15.0000	9.9704e-7	40.0000	7.2319e-6	7.2319e-6	2.6892e-3
47	16.0000	1.2551e-6	40.0000	8.6054e-6	8.6054e-6	2.9335e-3
48	17.0000	1.4678e-6	40.0000	1.0120e-5	1.0120e-5	3.1812e-3
49	18.0000	1.6998e-6	40.0000	1.1803e-5	1.1803e-5	3.4356e-3
50	19.0000	1.9509e-6	40.0000	1.3658e-5	1.3658e-5	3.6956e-3
51	20.0000	2.2287e-6	40.0000	1.5672e-5	1.5672e-5	3.9587e-3

V_S = 0 V

V_G = -30 V to 40 V

V_D = 40 V

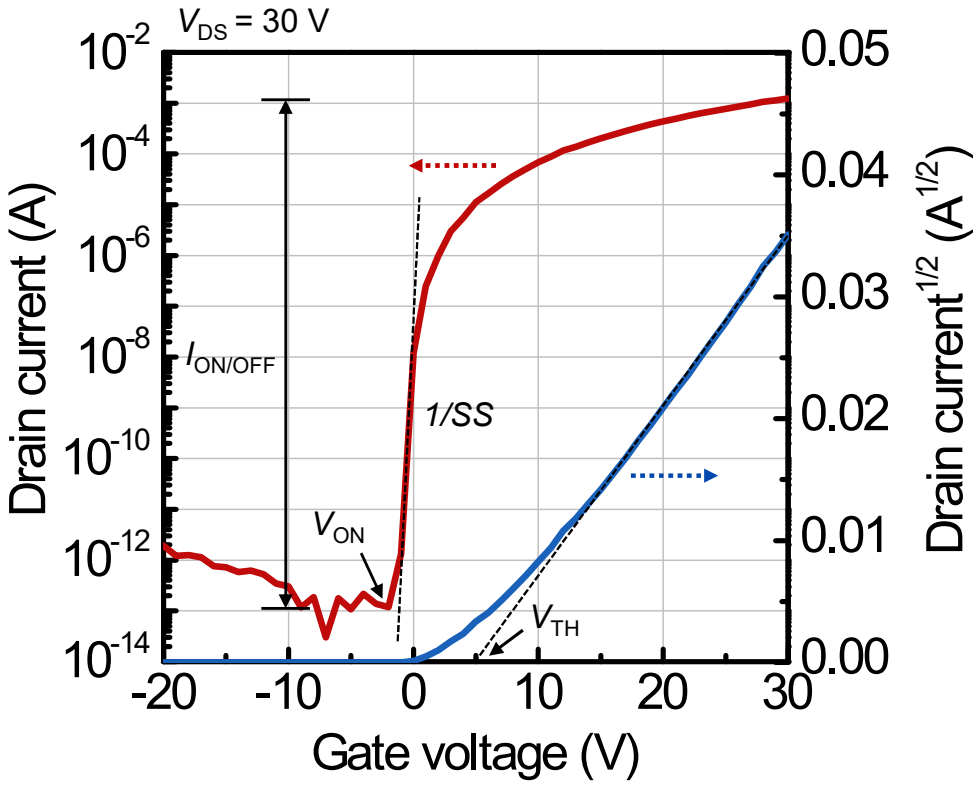
For n-type In_2O_3 TFTs

	1-VG	2-IG	3-VD	4-ID	5-|ID|	6-sqrt|ID|
1	-30.0000	-4.3325e-7	40.0000	2.1705e-7	2.1705e-7	4.6589e-4
2	-29.0000	-2.1314e-7	40.0000	1.4173e-7	1.4173e-7	3.7648e-4
3	-28.0000	-1.1400e-7	40.0000	5.2505e-8	5.2505e-8	2.2914e-4
4	-27.0000	-4.1587e-8	40.0000	3.1592e-8	3.1592e-8	1.7774e-4
5	-26.0000	-3.2616e-8	40.0000	2.5920e-8	2.5920e-8	1.6100e-4
6	-25.0000	-2.6971e-8	40.0000	2.2008e-8	2.2008e-8	1.4835e-4
7	-24.0000	-2.2757e-8	40.0000	1.8980e-8	1.8980e-8	1.3777e-4
8	-23.0000	-1.9595e-8	40.0000	1.6635e-8	1.6635e-8	1.2898e-4
9	-22.0000	-1.6971e-8	40.0000	1.4699e-8	1.4699e-8	1.2124e-4
10	-21.0000	-1.4937e-8	40.0000	1.3190e-8	1.3190e-8	1.1485e-4
11	-20.0000	-1.3291e-8	40.0000	1.1863e-8	1.1863e-8	1.0892e-4

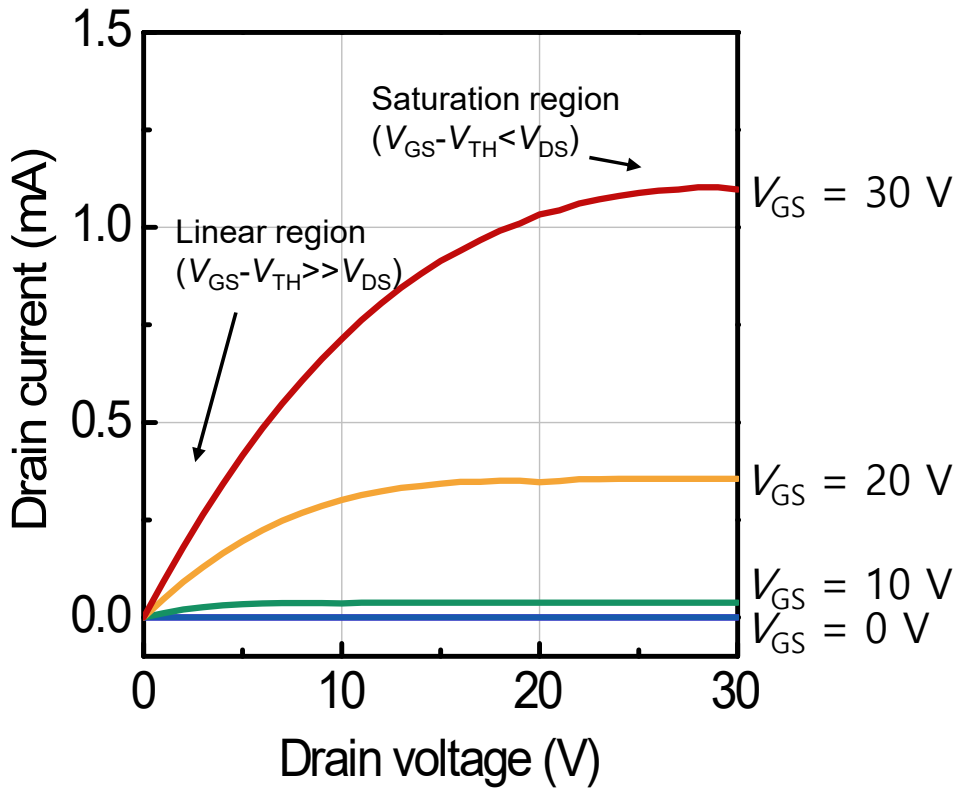
3. Device Measurement

3.3 Transfer and Output Characteristics

Transfer curve



Output curve



4. Analysis of Thin Films

4.1 Physical Ananlysis

OM
(Optical Microscope)



SEM
(Scanning Electron Microscope)



TEM
(Transmission Electron Microscope)

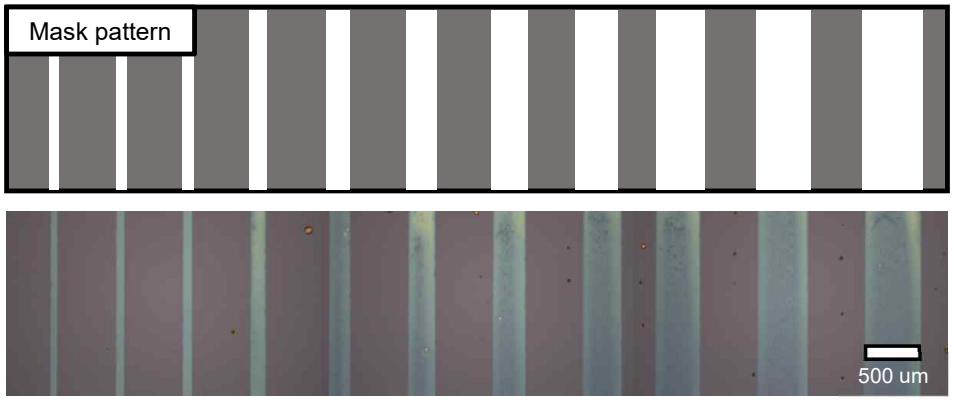


AFM
(Atomic Force Microscope)

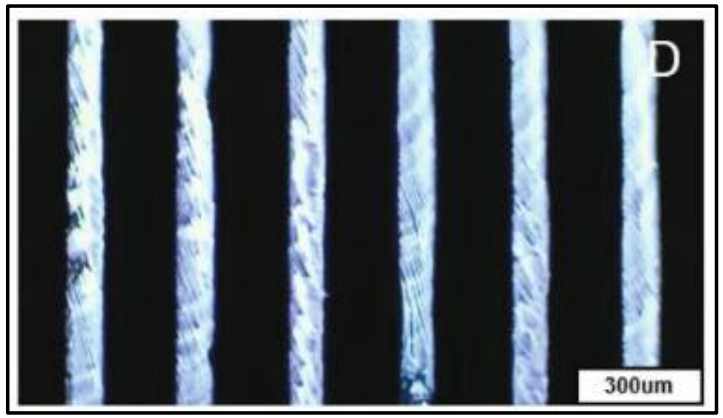


4. Analysis of Thin Films

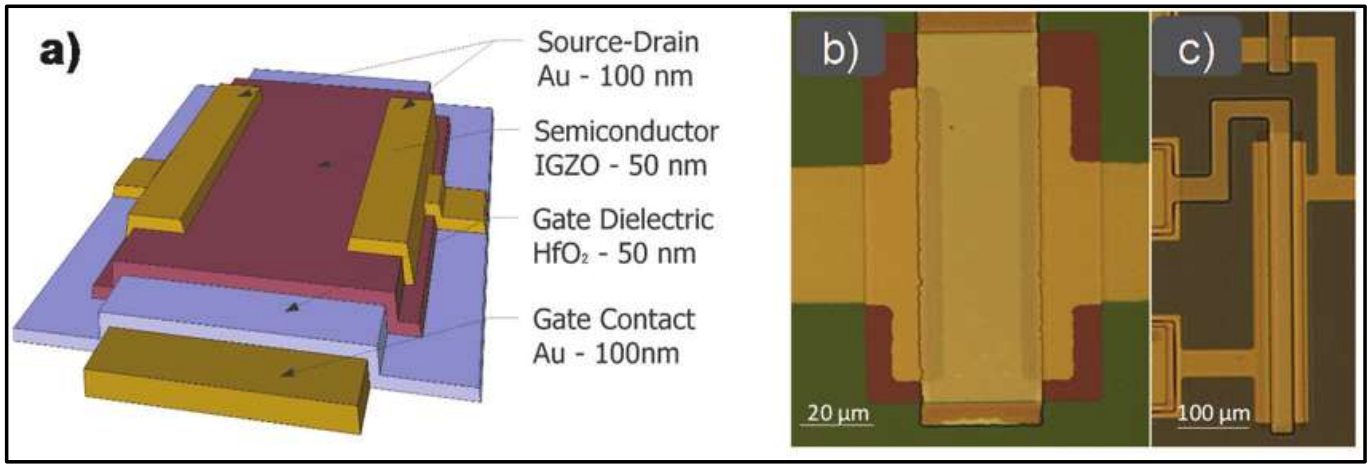
4.1 Physical Ananlysis : OM



Patterned IGZO semiconductor thin-films



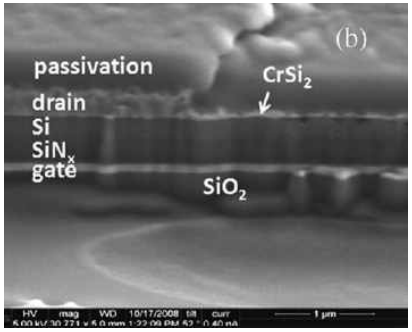
Patterned TIPS-PEN semiconductor thin-films

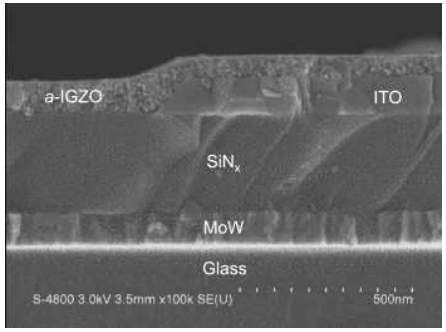


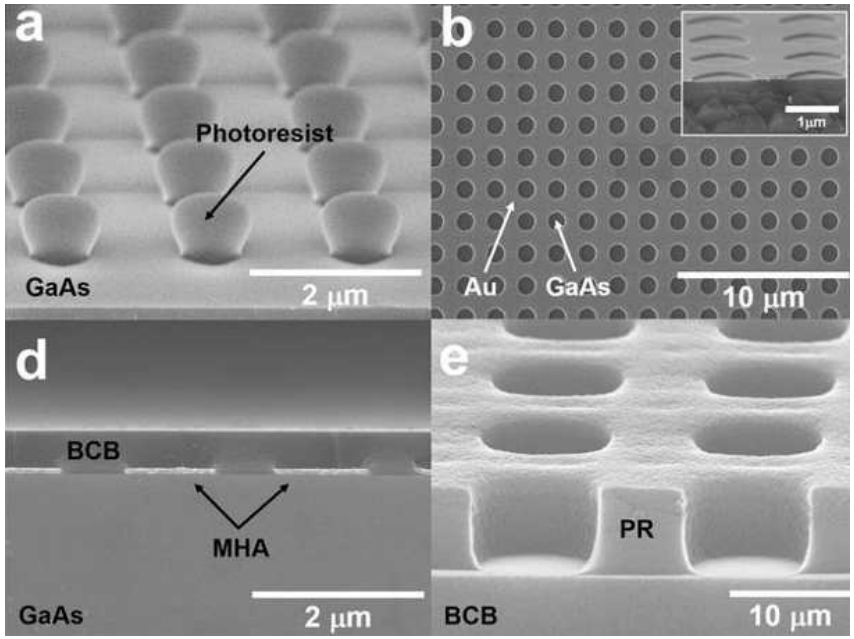
Patterned IGZO semiconductor TFTs

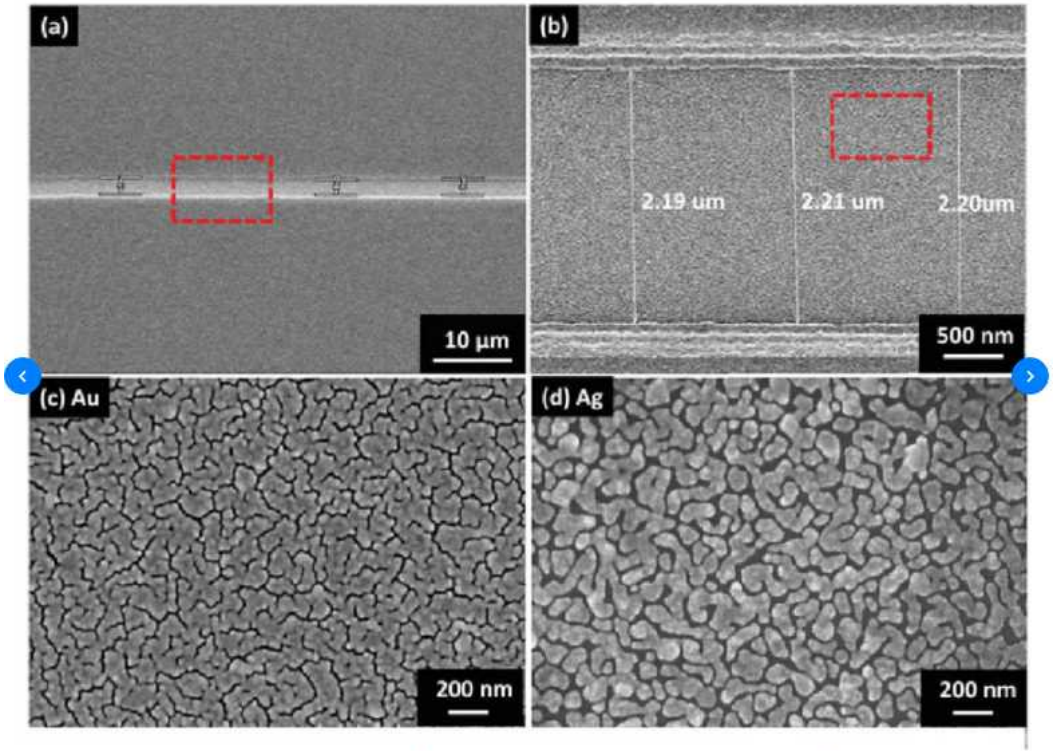
4. Analysis of Thin Films

4.1 Physical Ananlysis : SEM







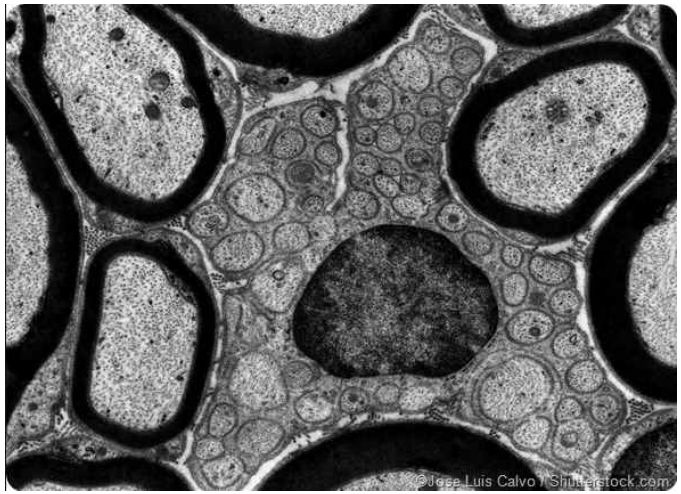


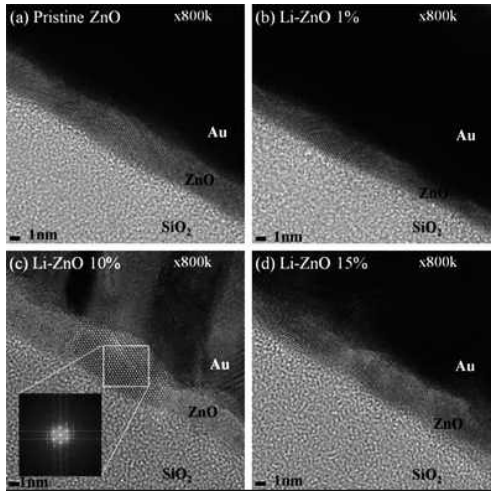
Scanning electron microscope (SEM) images of the Si samples after photolithography and metal deposition. (a) A 2 μm-wide line pattern on Si; (b) magnified image in the red-circled area in (a); (c) magnified image in the red-circled area in (c) where 10 nm-thick Au film was used as the catalyst; (d) magnified image in the red-circled area in (c) where 10 nm-thick Ag film was used as the catalyst.

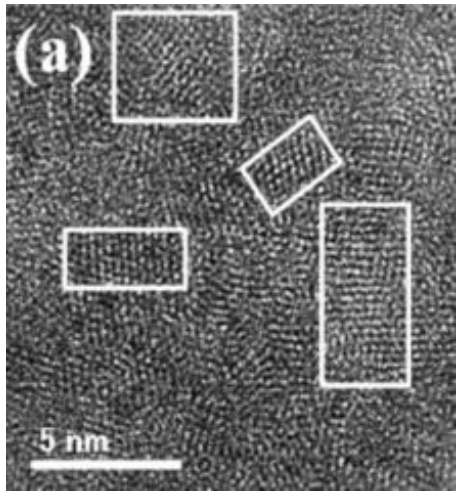
Semiconductor/conductor/insulator films, photoresist 등 다양한 박막 및 소자의 구조를 관측 가능

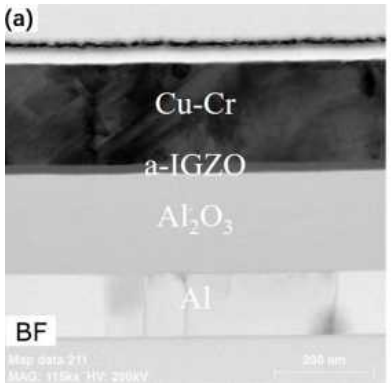
4. Analysis of Thin Films

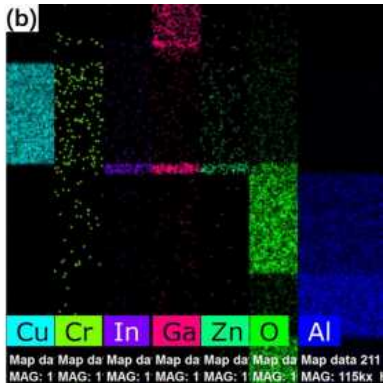
4.1 Physical Ananlysis : TEM

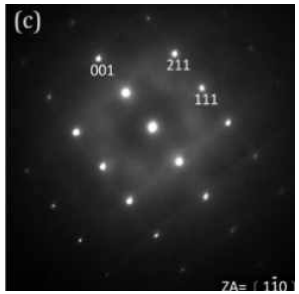


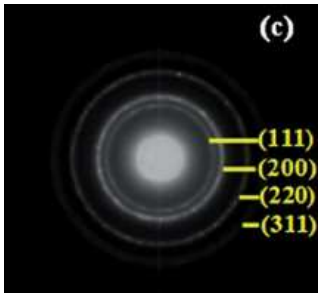


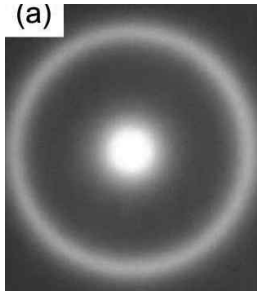












나노미터 단위의 세포, 박막, 원자 배열 등을 확인 가능.
원자별 분포, 박막의 결정성 등을 확인 가능.

4. Analysis of Thin Films

4.1 Physical Ananlysis : AFM

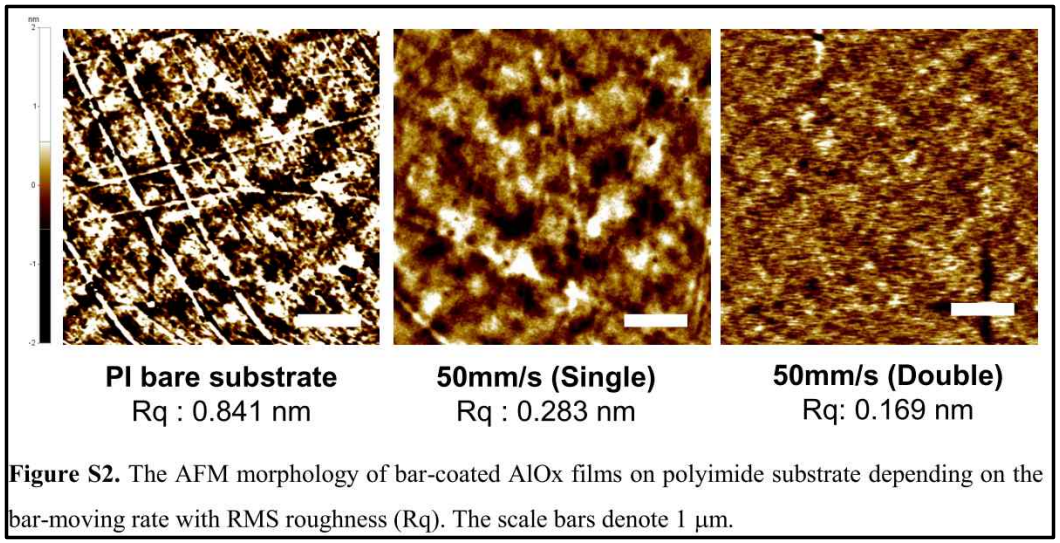
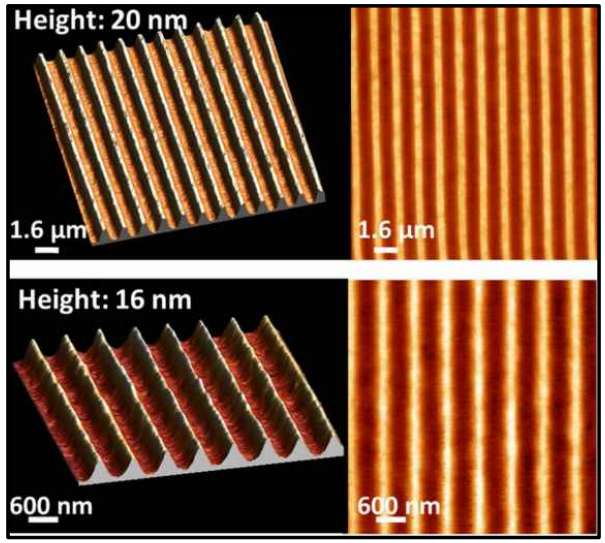
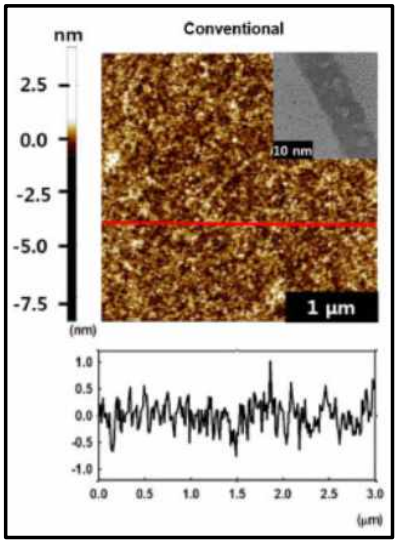


Figure S2. The AFM morphology of bar-coated AlOx films on polyimide substrate depending on the bar-moving rate with RMS roughness (Rq). The scale bars denote 1 μm.

PI 기판 및 AlOx high-k 절연층 박막



IGZO patterns

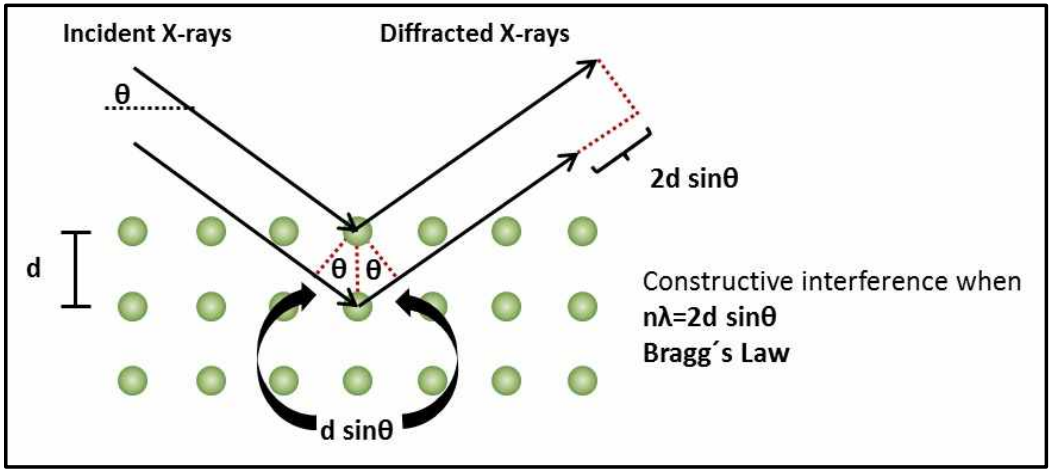


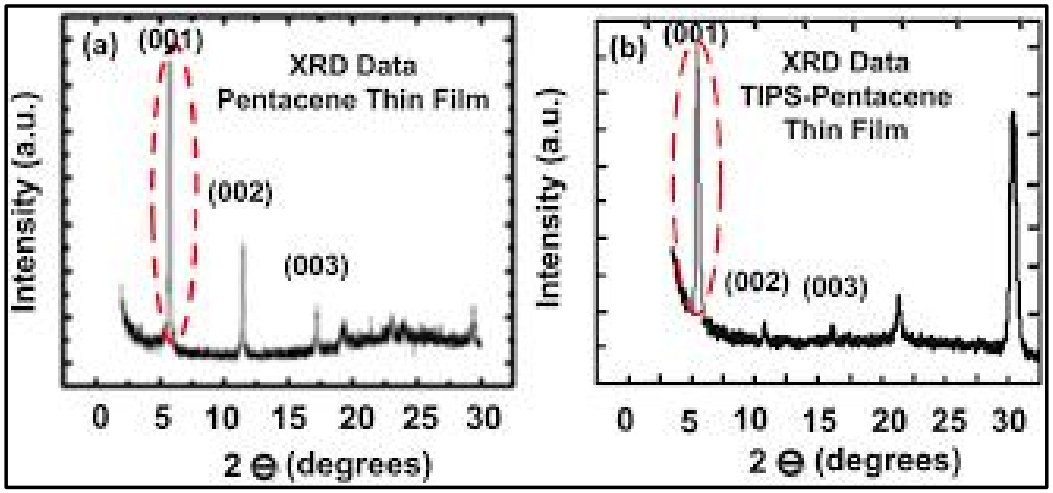
박막의 표면 구조를 nm단위로 확인 가능함.

4. Analysis of Thin Films

4.2 Chemical/Structural Ananlysis

XRD (X-ray diffraction)





원리: X선을 물질에 쏘아 나오는 반사파를 분석.

X선을 쓰는 이유: 파장이 짧아 박막 혹은 물질에 깊이 침투가 가능하여 분석이 가능함.

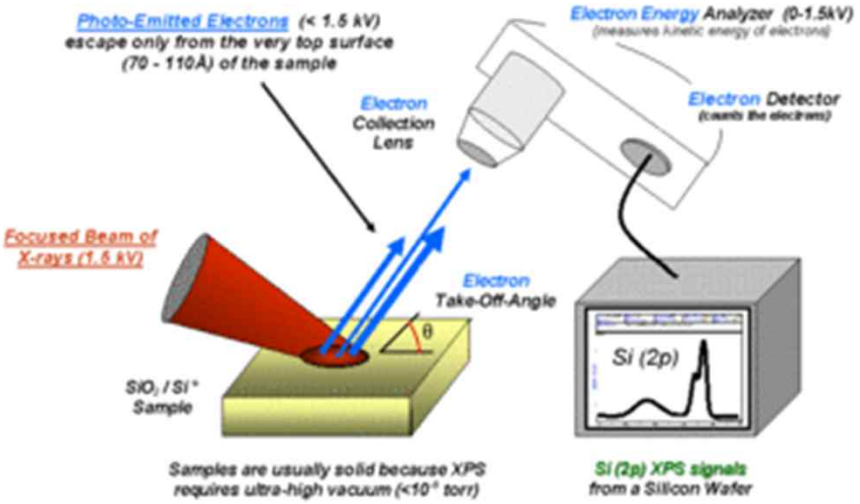
활용:

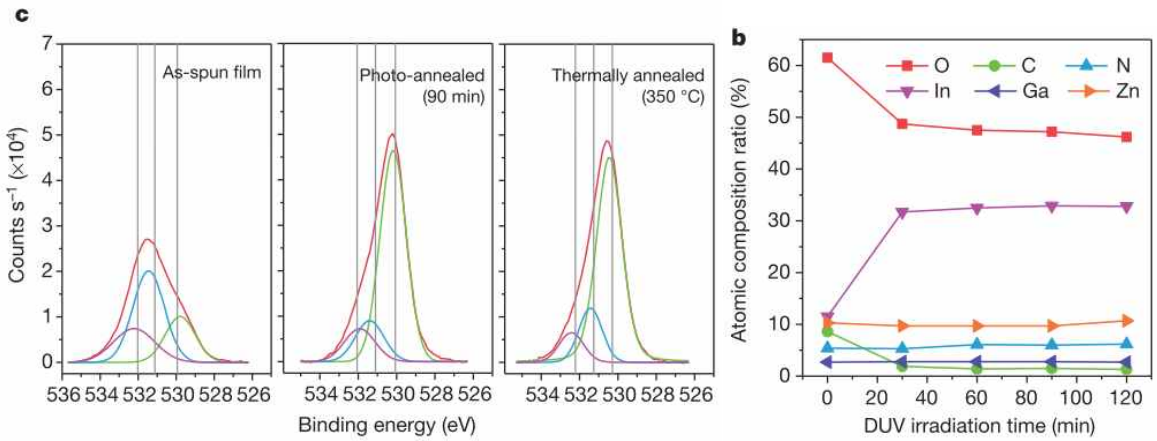
1) 물질의 내부 미세구조를 밝히는데 사용.
2) 결정 물질 속의 원자들의 배열, 상호거리, 고체의 물리적 성질을 밝힐 수 있음.
3) 비타민, 항생물질 등과 같은 복합적인 자연물의 구조를 밝히는데 이용.
4) 구성 성분을 모르는 임의의 시료의 성분을 회절패턴을 통해 밝힐 수 있음.

4. Analysis of Thin Films

4.2 Chemical/Structural Ananlysis

XPS (X-ray photoelectron spectroscopy)





원리: 광전 효과에 의해 방출된 전자의 운동 에너지를 측정하여 분석하는 표면 분석 방법.

광원으로서 X선을 사용하여 분석 시료를 이루고 있는 원소의 핵심부 전자를 방출 시킨 후, 방출된 전자의 운동에너지를 측정하여 시료를 구성하고 있는 원소를 찾을 수 있음.

활용:

1) 결합 에너지를 분석하여 확학 결합 상태 등에 대한 정보를 얻을 수 있음.

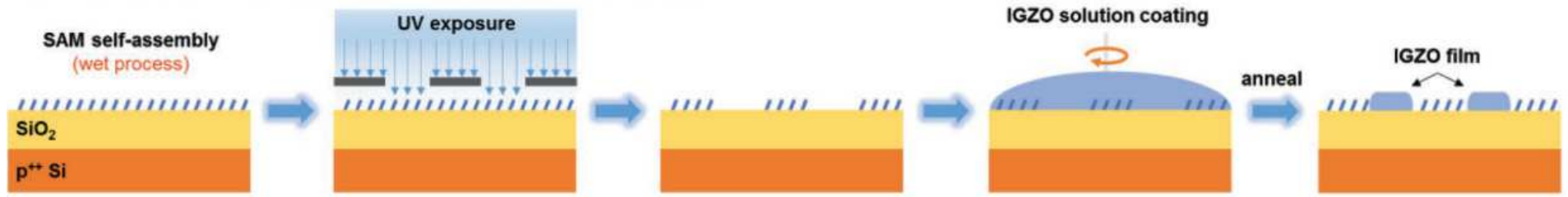
2) 박막, 코팅, 도금 등의 표 처리분야 뿐만 아니라 촉매, 반도체 등 여러 분야의 다양한 재료에 대한 표면 분석 가능.

5. Patterning Technologies for Solution-Processed Oxide TFTs

5.1 Expeditious Solution-Free Surface Energy Patterning

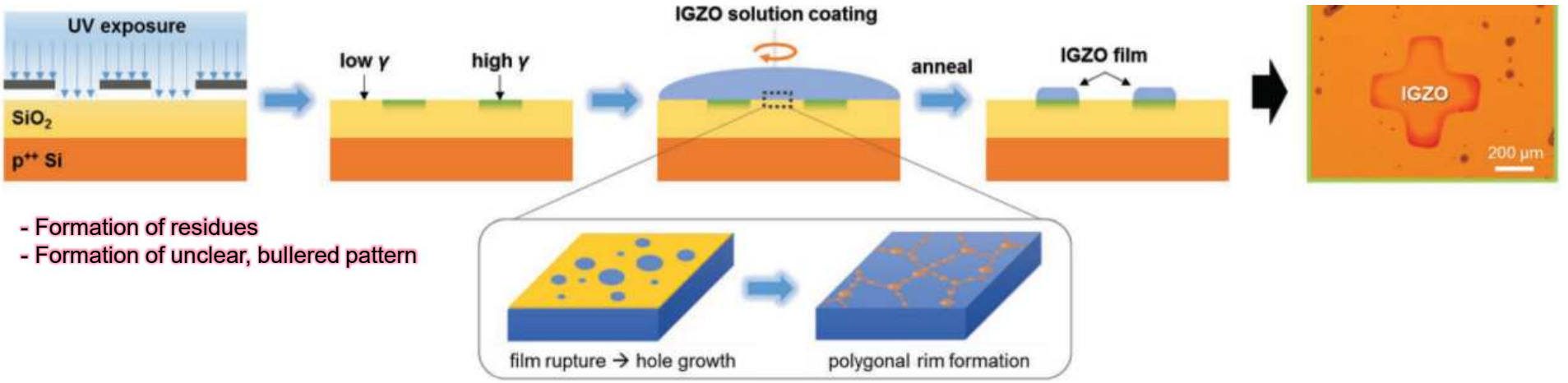
Introduction : Surface Energy Patterning of Sol-Gel Oxide Thin Films

Chemical Solution(Self-Assembled Monolayer; SAM)-Based Surface Energy Patterning



- Requires long time (1 – 8 h) for SAM deposition
- Wet chemistry (Solution waste, toxicity, hindering mass production)

Solution-Free Surface Energy Patterning



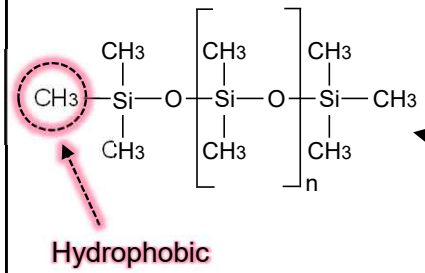
- Formation of residues
- Formation of unclear, bullered pattern

5. Patterning Technologies for Solution-Processed Oxide TFTs

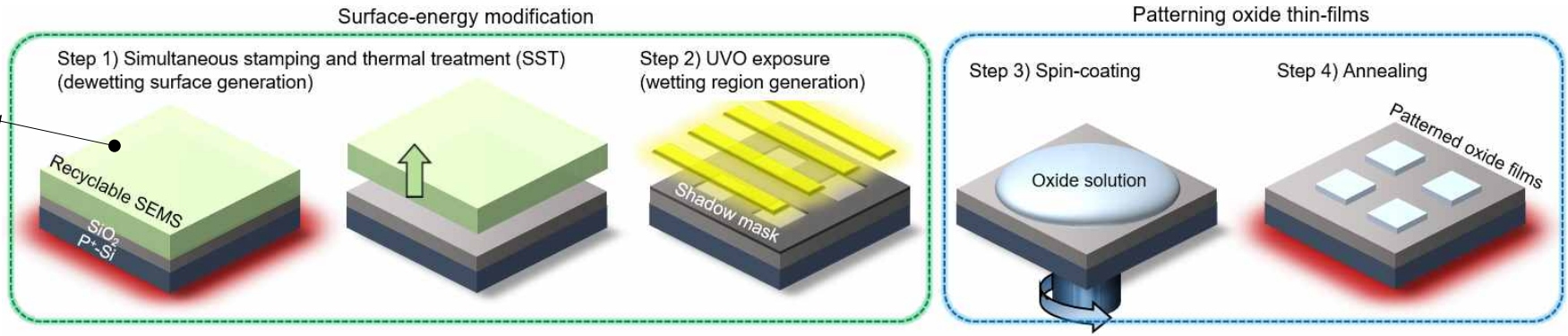
5.1 Expeditious Solution-Free Surface Energy Patterning

Fabrication Process: Sustainable Solution-Free Surface Energy Patterning

Chemical structure of PDMS

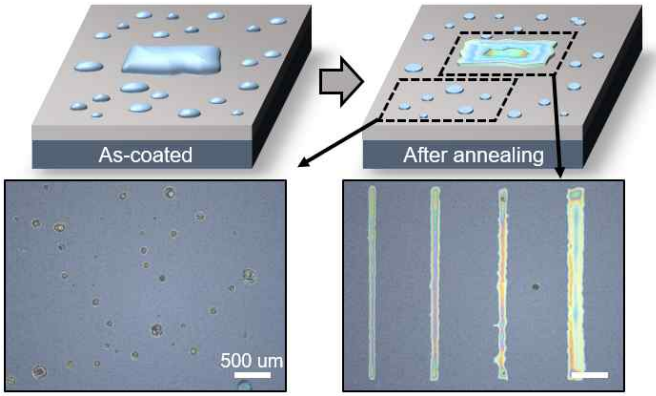


Surface Energy Patterning (SEP) Process

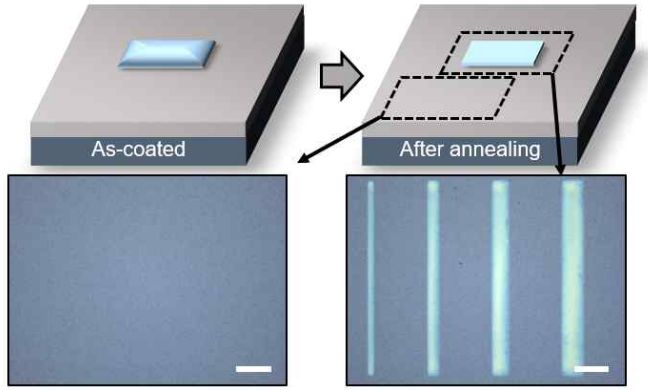


Effect of SST

Patterning w/o SST



Patterning w/ SST



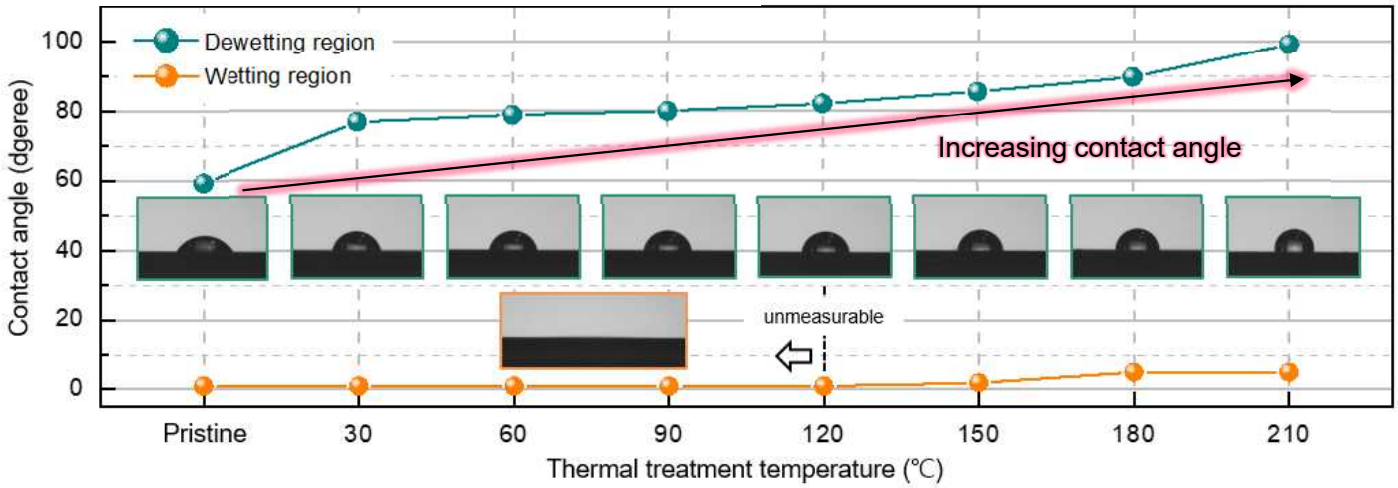
- Without SST:
  - smudgy and uneven IGZO patterns
  - undesirable dot-shaped residues and polygonal rims

- With SST:
  - clear and flat IGZO patterns without any residues (due to the sufficient hydrophobicity and surface energy difference)

5. Patterning Technologies for Solution-Processed Oxide TFTs

5.1 Expeditious Solution-Free Surface Energy Patterning

Thermal Effect on Surface Energy in SST



Young's Equation

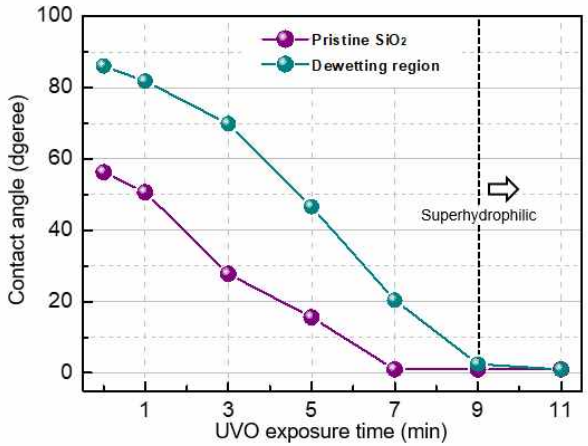
$$\gamma_{SL} = \gamma_S - \gamma_L \cos\theta$$

$$\gamma_{SL} = \gamma_S + \gamma_L - 2(\gamma_S\gamma_L)^{\frac{1}{2}}$$

$$\gamma_S = \frac{\gamma_L}{4}(1+\cos\theta)^2$$

γ_{SL} : interfacial free energy between a solid and liquid
γ_S : surface free energy of a liquid
γ_L : surface free energy of a solid
θ : contact angle

UV Effect on Surface Energy



Chapman Cycle

Formation of Ozone

$$O_2 \xrightarrow{\lambda<240\ nm} 2O\bullet$$

$$O\bullet + O_2 \rightarrow O_3$$

Destruction of Ozone

$$O_3 \xrightarrow{\lambda<300\ nm} O_2 + O\bullet$$

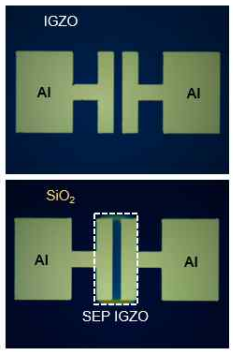
$$O + O_3 \rightarrow 2O_2$$

Formation of Hydrophilic surface

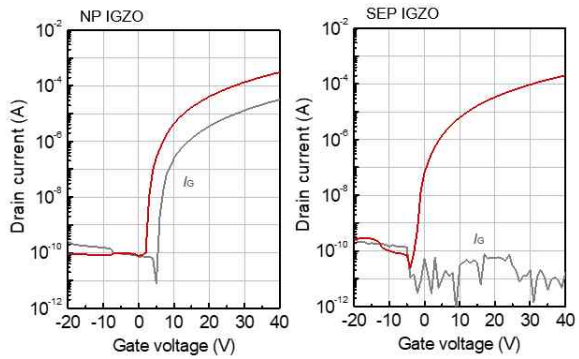
1) $Si - CH_3$
2) $Si -$
3) $Si - O\bullet$
4) $Si - OH$

Electrical Properties of SEP IGZO TFTs

Optical Microscope



Transfer Characteristics

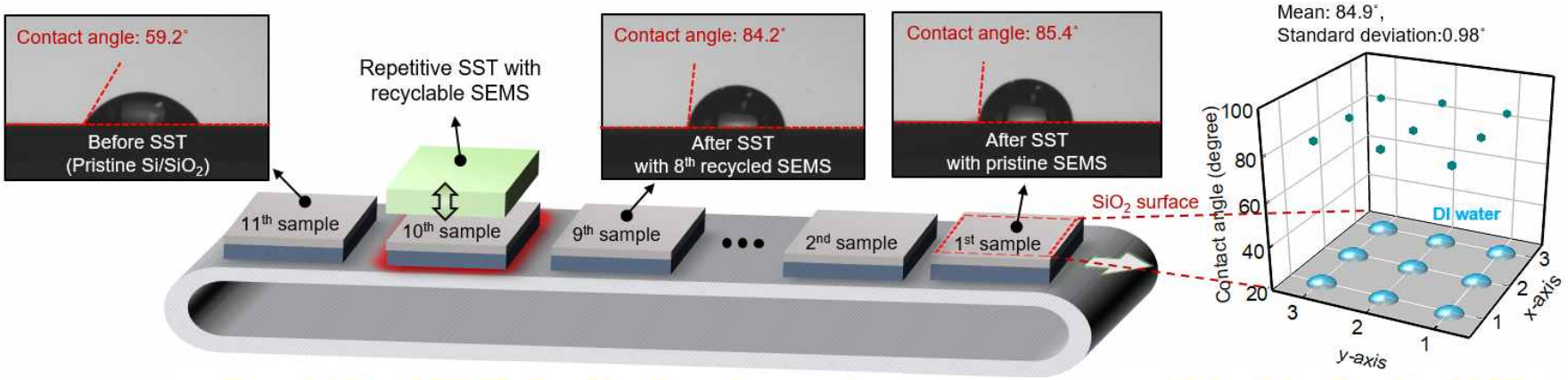


5. Patterning Technologies for Solution-Processed Oxide TFTs

5.1 Expeditious Solution-Free Surface Energy Patterning

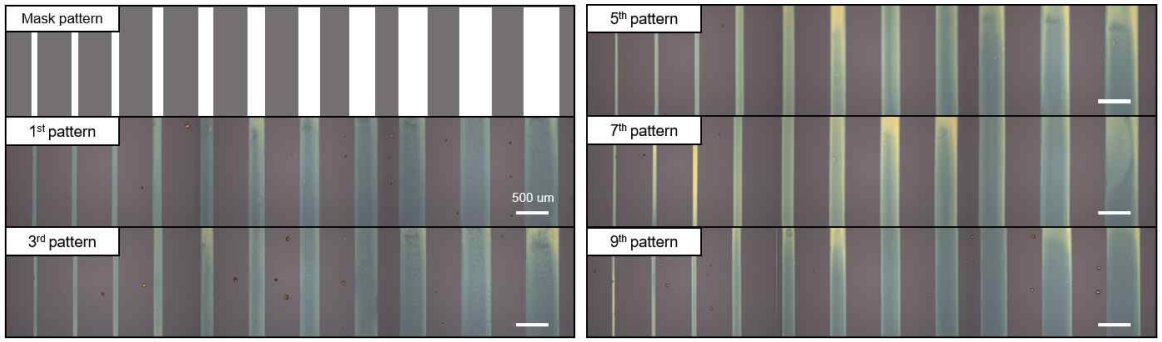
Recyclability of SEMS

Surface Energy

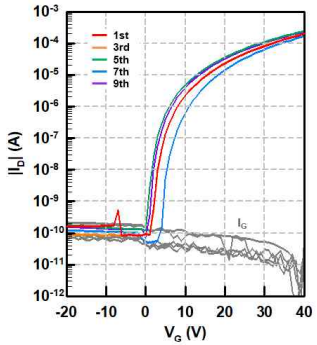


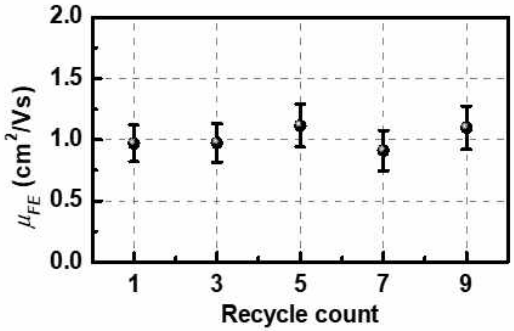
- Recyclability of SEMS: Eco-friendly and economic process
- High arial uniformity of SST

SEP IGZO Patterns



Transfer Characteristics of SEP IGZO TFTs





- Clear IGZO pattern and uniform electrical properties of SEP IGZO TFTs were obtained although using only one SEMS.

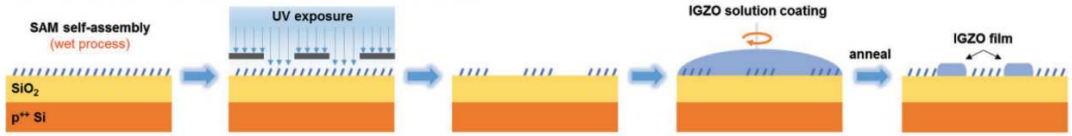
5. Patterning Technologies for Solution-Processed Oxide TFTs

5.2 Water Etchant-Based Photopatterning for Low-Defect Oxide Films

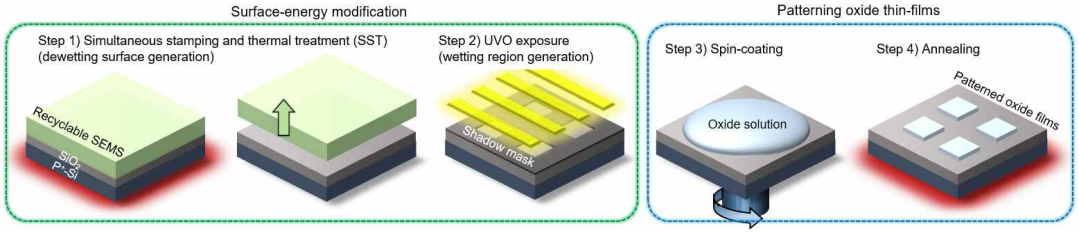
Introduction : Patterning Methods of Sol-Gel Oxide Thin Films

Additive approach : Surface energy patterning

Chemical solution-based surface energy patterning



Solution-free surface energy patterning

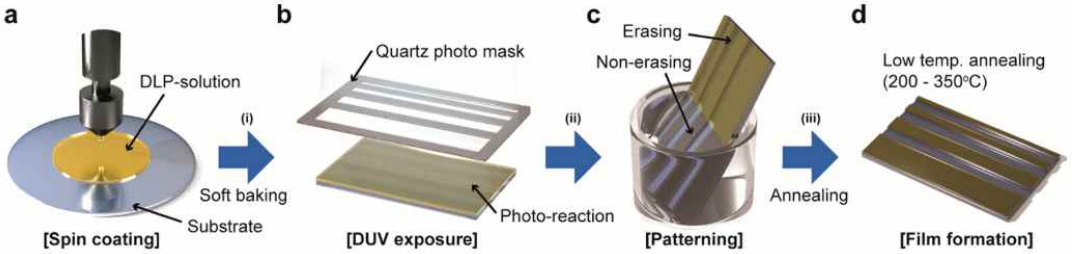


1. Advantages
- Facile method
- Material-independent method

2. Disadvantages
- Low-resolution
- Nonuniform morphology
- Inducing low electrical properties and reliability

Subtractive approach : Photopatterning

Photopatterning



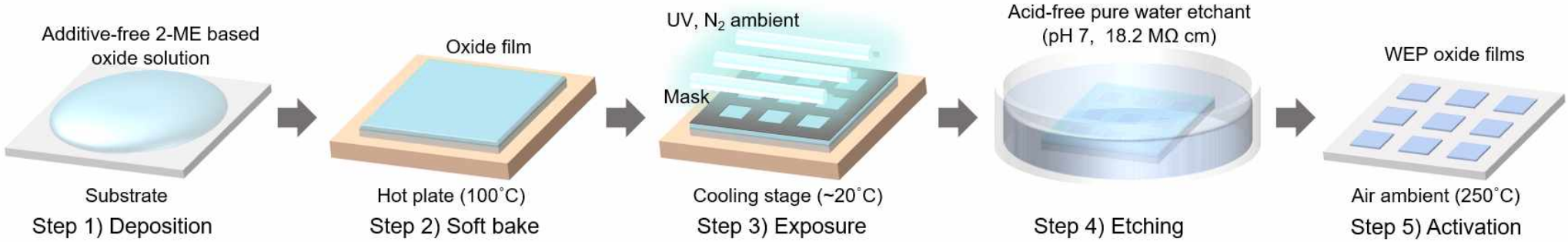
1. Advantages
- Facile method
- High-resolution
- Uniform morphology
- Photoresist-free method
- No degradation of electrical properties

2. Disadvantages
- Requires photoreactive materials
- Toxic acid solution or solvent for etchant

5. Patterning Technologies for Solution-Processed Oxide TFTs

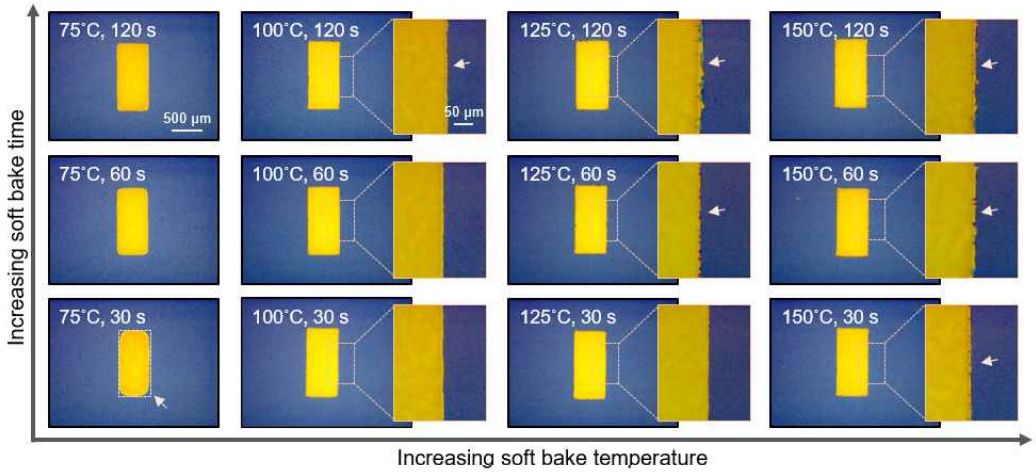
5.2 Water Etchant-Based Photopatterning for Low-Defect Oxide Films

Experimental

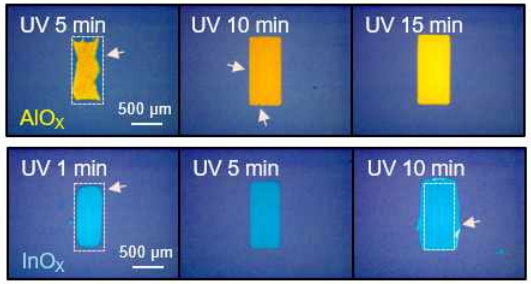


Patterning Parameters Effect on Pattern Morphology

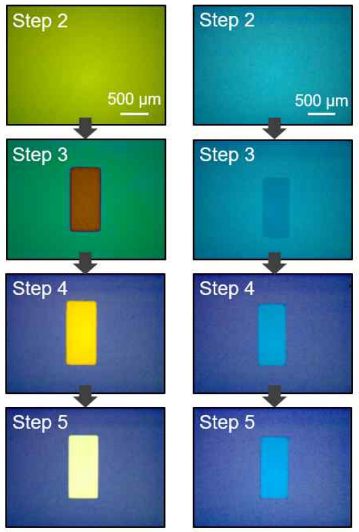
Soft Bake Time and Temperature



UV Time



AlOx and InOx in Each Step



- Clear oxide patterns were obtained by controlling key parameters in water etchant-based photopatterning.

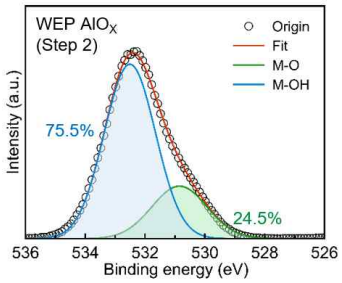
5. Patterning Technologies for Solution-Processed Oxide TFTs

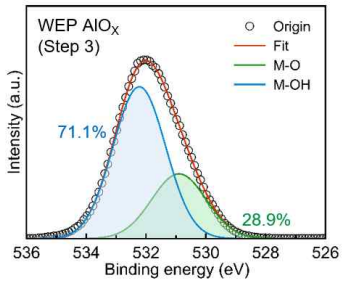
5.2 Water Etchant-Based Photopatterning for Low-Defect Oxide Films

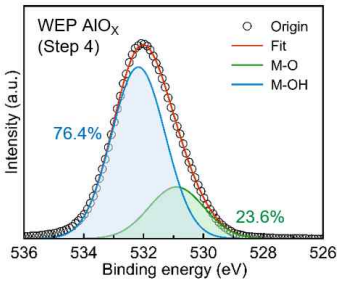
Chemical Composition of WEP AlO*x*

XPS O 1s and N 1s spectra

* Step 2: soft bake, Step 3: UV exposure, Step 4: etching, Step 5: activation



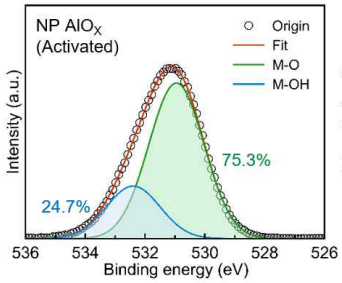


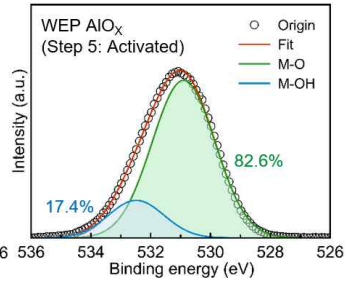


- Hydroysis
  MX+ROH → M-OR
  MX+H_2O → M-OH

- Cleavage of alkoxides
  M-OH + M-OR → M-O-M
- Decomposition of nitrate ligands
  NO_3^- + hv → NO_2 + OH
  MX+H_2O or ROH → M-OH or M-OR

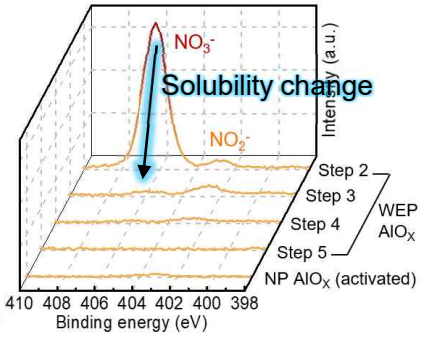
- Hydroysis
  MX+H_2O → M-OH





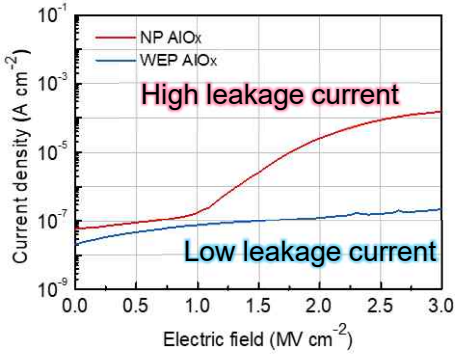
- Condensation & Densification
  Low M-O, High M-OH

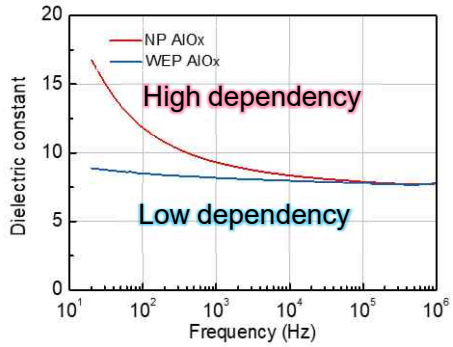
- Condensation & Densification
  High M-O, Low M-OH

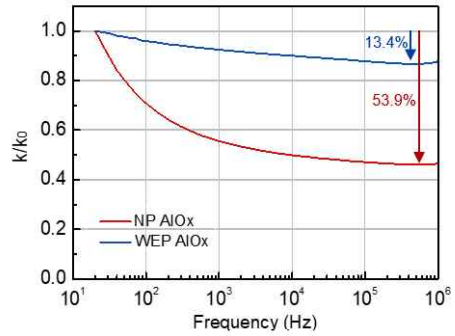


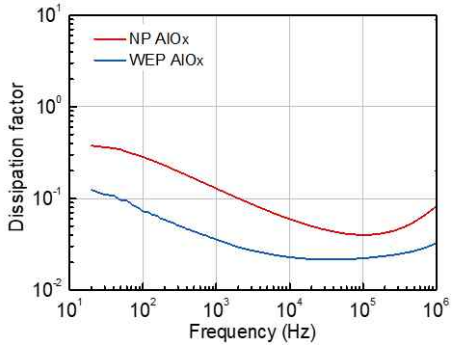
Electrical Properties of WEP AlO*x*

AlOx-based MIM Capacitors









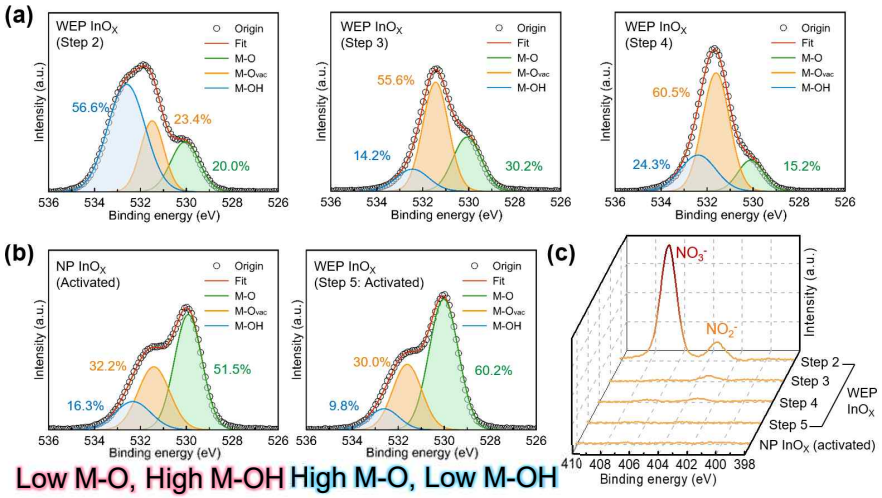
- WEP AlO*x* capacitor
- Sufficiently low leakage current density
  (2.2 × 10^{-7} A cm^{-2} at 3 MV cm^{-1})
- Frequency-stable dielectric constant
  (within 7-8, decrease 13.4%)

5. Patterning Technologies for Solution-Processed Oxide TFTs

5.2 Water Etchant-Based Photopatterning for Low-Defect Oxide Films

Chemical Composition of WEP InO*x*

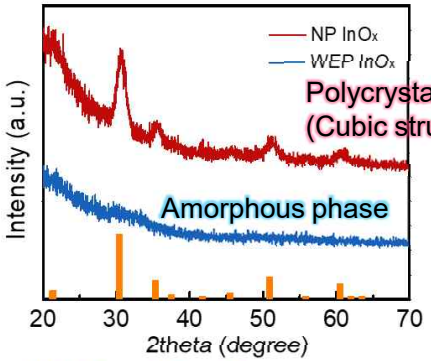
XPS O 1s and N 1s spectra



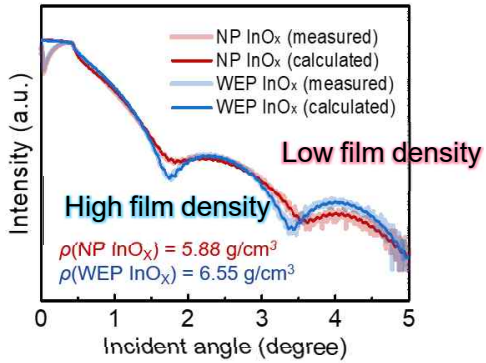
Low M-O, High M-OH High M-O, Low M-OH

Physical Properties of WEP InO*x*

GIXRD Analysis



XRR Analysis



- Water etchant:
  - Suppression of chemical defects (dense M-O-M networks)
  - Suppression of crystallization and physical defects

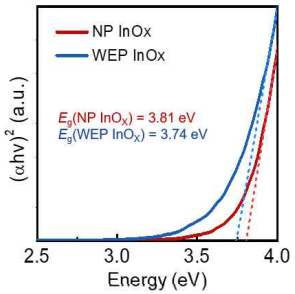
Energy Level of WEP InO*x*

Tauc Equation

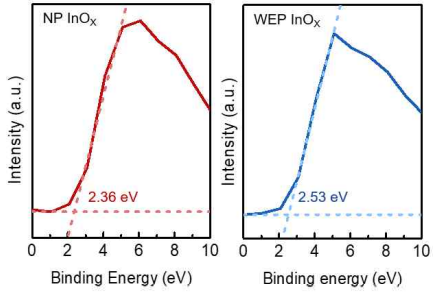
$(\alpha h\nu)^2 = A(h\nu - E_g)$

α : absorption coefficient
$h\nu$: photon energy
E_g : bandgap

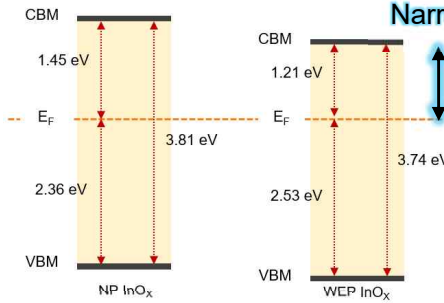
Optical Bandgap



Valence Band Offset



Energy Level Diagram



Narrower gap between E_F and CBM (1.21 eV)

Lower E_g (3.74 eV)

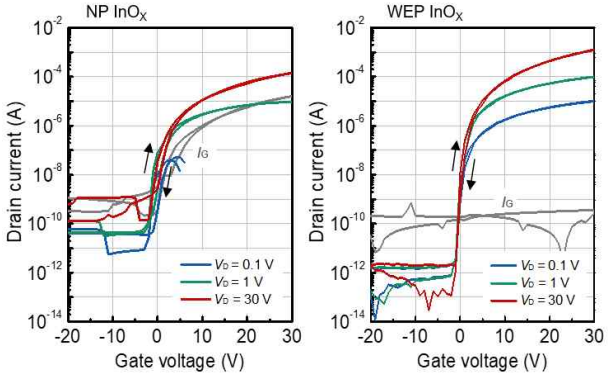
- The lower E_g, narrow gap between the Fermi level and CBM of WEP InO*x* → higher carrier density

5. Patterning Technologies for Solution-Processed Oxide TFTs

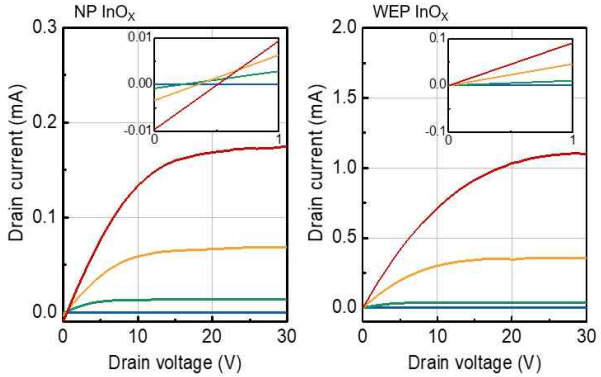
5.2 Water Etchant-Based Photopatterning for Low-Defect Oxide Films

Electrical Properties of WEP InO*x* TFTs

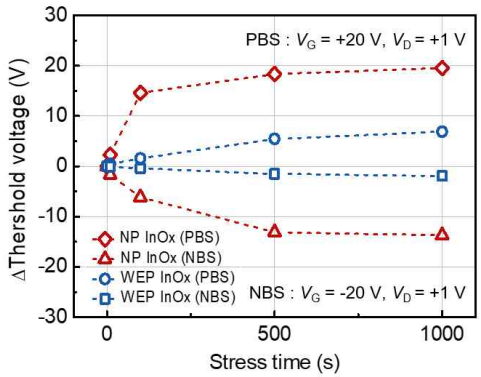
Transfer Characteristics



Output Characteristics



Gate Bias Stability



- **NP InO*x* TFTs:**
  - μ_{sat} = 0.80 cm^2 V^{-1} s^{-1}
  - *SS* = 0.89 V dec^{-1}
  - $I_{on/off}$ = 10^6
- **WEP InO*x* TFTs:**
  - μ_{sat} = 12.04 cm^2 V^{-1} s^{-1}
  - *SS* = 0.37 V dec^{-1}
  - $I_{on/off}$ = 10^9

Charge Transport Mechanism

Conventional MOSFET Theory

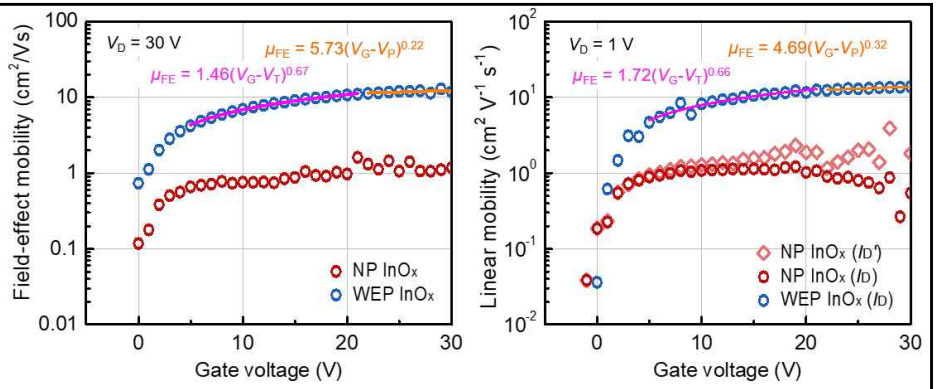
$$\mu_{h} = \frac{L}{WC_iV_D}\cdot\frac{\partial I_{D,h}}{\partial V_G} \qquad \mu_{sat} = \frac{2L}{WC_i}\cdot\frac{\partial^2 I_{D,sat}}{\partial V_G^2}$$

Multiple Trapping and Release (MTR) Model

$$\mu_{ef} = \mu_0\left(1+\frac{T}{Tc}\right)\left(\frac{C_i(V_G-V_0)}{qN_T}\right)^{\left(\frac{Tc}{T}-2\right)}$$

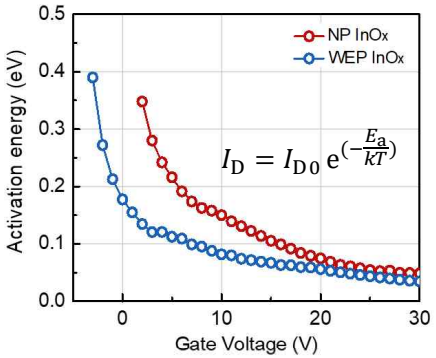
Compact Power Law Model

$$\mu_{FE} = K(V_G - V_{T,P})^{\gamma} \rightarrow \gamma \sim 0.7\text{: Trap Limited Conduction (TLC)},\ \gamma \sim 0.1\text{: Percolation Conduction (PC)}$$



- The dominant electron conduction mechanism of WEP InO*x* TFTs:
  Transited from the TLC to PC as the gate electric field increased.

Activation Energy

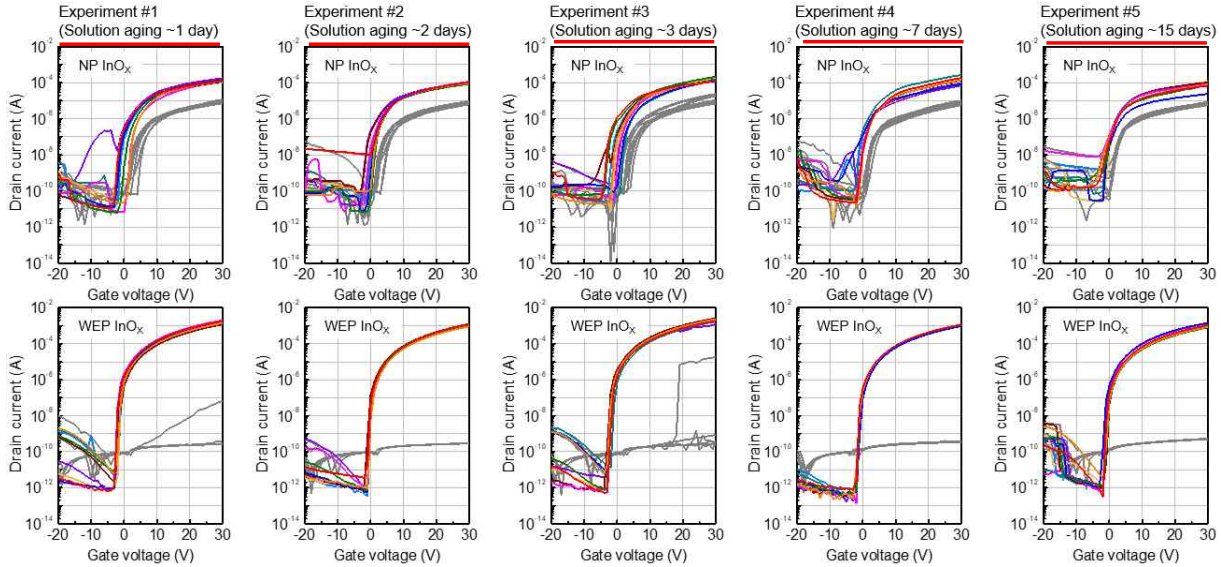


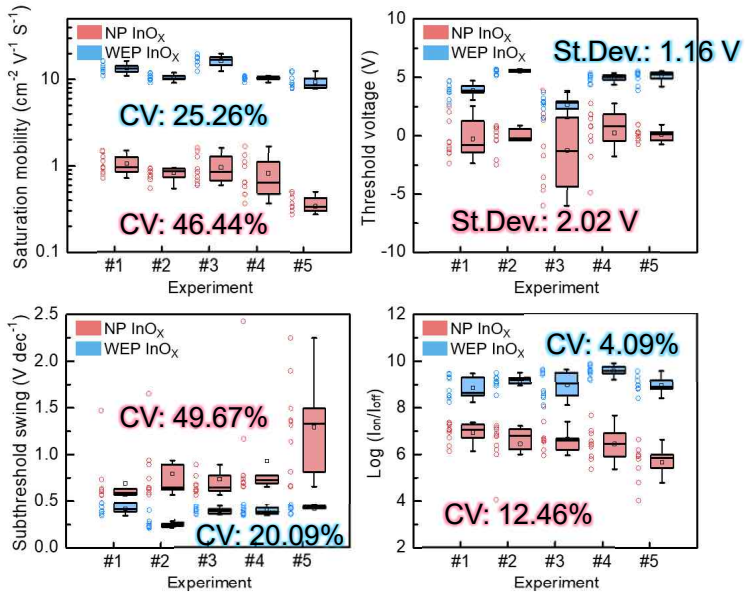
- Low E_a:
  - Low localized tail states (traps)
  - Weak mobility dependence on temp.

5. Patterning Technologies for Solution-Processed Oxide TFTs

5.2 Water Etchant-Based Photopatterning for Low-Defect Oxide Films

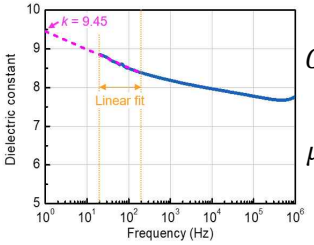
Device-to-Device Uniformity of InO*x* TFTs





- WEP InO*x* TFTs (5 repetitive experiments, 10 devices per time): 15-fold higher μ_{avg}, excellent device-to-device uniformity, solution aging stability

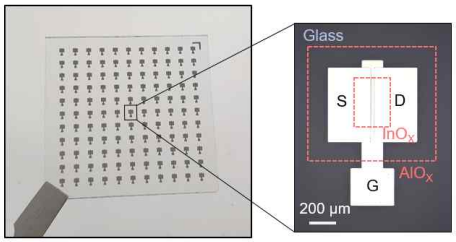
3-V Operating InO*x*/AlO*x* TFTs



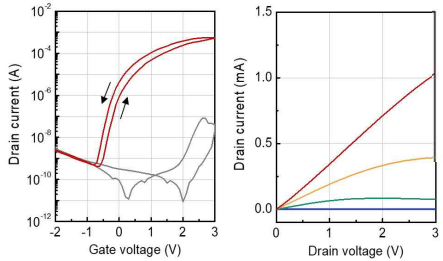
$$C = \frac{k\varepsilon_0 A}{d}$$

$$\mu_{sat} = \frac{2L}{WC_i} \cdot \frac{\partial^2 I_{D,sat}}{\partial V_G^2}$$

Fabricated InO*x*/AlO*x* TFTs



Transfer/Output Characteristics



- Low processing temperature of 250˚C
- Low voltage operation of 3 V
- High mobility of 24.77 $cm^2\ V^{-1}\ s^{-1}$